GENERAL SCIENCE GRADE 10: NUTRITION IN ANIMALS

CBE Phenomenon-Driven Lesson Sequence — Sub-Strand 1.3 (8 Lessons)

SUB-STRAND OVERVIEW	
Grade Level	10
Subject	General Science
Strand	Strand 1.0: Life Science
Sub-Strand	Sub-Strand 1.3: Nutrition in Animals
Total Duration	8 lessons × 40 minutes = 320 minutes total
Sub-Strand Content	– Chemical and mechanical digestion of foods – Adaptations of parts of the human digestive system – Food tests (test for reducing sugars, starch, proteins and fats)
Learning Outcomes	By the end of the sub-strand, the learner should be able to: a) describe the digestion of different types of food in each region of the digestive system, b) explain the adaptations of parts of the human digestive system, c) perform experiments to determine presence of different nutrients in food, d) appreciate the role played by different parts of the digestive system.
Core Competencies	– Digital Literacy: as the learner retrieves and saves information from non-print sources on the enzymes used to break down each type of food at different regions of the digestive system and shares the information with peers using digital presentation modes. – Creativity and Imagination: as the learner uses locally available materials to make a model of the human digestive system.
Core Values	– Respect: as the learner considers others' opinions while discussing the adaptations of the human digestive system. – Unity: as the learner works harmoniously when carrying out experiments on food tests.
Science & Engineering Practices	– Asking Questions and Defining Problems: as learners generate and refine questions about why different regions of the digestive system are structured differently when observing the anchoring phenomenon. – Planning and Carrying Out Investigations: as learners design and conduct food-test experiments (Benedict's test, iodine test, Biuret test, ethanol emulsion test) to determine nutrient presence in Kenyan foods such as ugali, githeri and sukuma wiki. – Analyzing and Interpreting Data: as learners record and interpret colour-change results from food tests, comparing findings across different food samples. – Constructing Explanations: as learners use evidence from experiments and digital research to explain how each region of the alimentary canal is adapted to its digestive function. – Developing and Using Models: as learners build and annotate a physical model of the human digestive system using locally available materials, linking structural adaptations to functions.

Pertinent & Contemporary Issues (PCIs)	– Safety and Security: as the learner observes safety precautions as they carry out experiments on food tests.
Career Connections	– Nutritionist / Dietitian: understanding how different nutrients are digested and absorbed guides dietary planning for Kenyan communities. – Medical Doctor / Clinical Officer: knowledge of digestive system anatomy and pathology underpins diagnosis and treatment of gastrointestinal conditions. – Food Scientist / Food Technologist: food-test techniques and nutrient analysis are core skills in quality control of Kenyan food products. – Public Health Officer: understanding digestion and malnutrition informs community health programmes such as school feeding initiatives. – Biochemist / Laboratory Technician: performing and interpreting chemical food tests is foundational to biochemistry laboratory work. – Agricultural Extension Officer: linking crop nutrient profiles (e.g. maize, beans) to human digestive needs supports food-security advice.
Focus for Lessons	This unit focuses on how the human digestive system breaks down food — both mechanically and chemically — and how each region is structurally adapted to its specific digestive function. Learners investigate nutrient composition through hands-on food tests using common Kenyan foods (ugali, githeri, nyama choma, sukuma wiki, mandazi), and build cumulative models of the alimentary canal that link structure to function. The unit employs a phenomenon-driven, inquiry-based approach in which learners move from personal puzzlement about a familiar meal through structured investigations to evidence-based explanations and model building.
Driving Question / Key Inquiry Questions	DRIVING QUESTION: How is the human digestive system adapted to break down and absorb the nutrients in a typical Kenyan meal — and what happens when those adaptations fail? KICD KEY INQUIRY QUESTION: 1. How is the human digestive system adapted to its functions?
Anchoring Phenomenon	ANCHORING PHENOMENON — "The Ugali and Sukuma Wiki Puzzle": A standard Kenyan school lunch of ugali (maize meal), sukuma wiki (kale), and a small portion of nyama choma (grilled meat) is placed before learners. Students observe that the meal contains visibly different substances — starchy, fibrous, and protein-rich — yet within hours the body has apparently extracted energy and nutrients from all of it, leaving only waste. What is puzzling is that a single, uniform-looking lump of ugali and a leaf of sukuma wiki enter the mouth as solid, complex matter, yet the small intestine — a tube barely 3 cm wide — somehow absorbs the nutrients into the bloodstream without any visible machinery. Learners cannot explain: (1) how the body "knows" which chemical to release at each stage, (2) why digestion takes different amounts of time for carbohydrates versus proteins, and (3) how the body avoids digesting itself. This everyday, culturally familiar meal creates genuine cognitive dissonance that motivates the entire unit's investigation.
Storyline Thread	Lesson 1 — PREDICT | Anchoring Phenomenon Launch: Teacher presents the "Ugali and Sukuma Wiki Puzzle" — a full Kenyan school lunch is displayed (or photographed). Learners individually write predictions about what happens to each food component (ugali, sukuma wiki, nyama choma) after swallowing. Class constructs an initial Driving Question Board (DQB) listing what they notice, what they wonder, and what they already think they know. Learners sketch a first draft "journey map" of food through the body from memory alone. Lesson 2 — OBSERVE | Mechanical vs. Chemical Digestion in the Mouth and Stomach: Learners watch an animation/simulation

	of food being chewed and mixed with saliva (mechanical digestion) and then churned in the stomach (chemical digestion with pepsin and HCl). Teacher demonstrates the mandazi-in-the-mouth phenomenon live. Learners record observations and compare the two types of digestion, identifying the roles of teeth, tongue, salivary glands, and stomach wall. DQB is updated with new questions arising from the animation. Lesson 3 — OBSERVE | Enzymes and Regions of the Alimentary Canal: Learners use print/digital media (Kenya Education Cloud or printed resource sheets) to research the specific enzymes active in each region — mouth (salivary amylase), stomach (pepsin, rennin), small intestine (pancreatic amylase, lipase, trypsin, intestinal peptidases) — and the type of food each breaks down. Groups create a region-by-region enzyme summary chart and present findings digitally or on manila paper. Teacher cold-calls 3 groups to explain one enzyme each. DQB updated. Lesson 4 — OBSERVE & EXPLAIN | Food Tests Practical (Part 1 — Starch and Reducing Sugars): Learners carry out the iodine test for starch and Benedict's test for reducing sugars using samples of ugali, ripe banana, raw maize flour, sugar solution, and sukuma wiki extract. Teacher gives step-by-step procedural guidance (Benedict's test: 2 cm^3 test solution + equal Benedict's reagent → hot water bath → observe colour change). Learners record all colour changes in results tables, identify which Kenyan foods contain starch vs. reducing sugars, and link findings back to which enzymes would act on these substrates in the body. Lesson 5 — OBSERVE & EXPLAIN | Food Tests Practical (Part 2 — Proteins and Fats): Learners perform the Biuret test for proteins (using nyama choma extract, bean broth, milk) and the ethanol emulsion test for fats (using butter, cooking oil, sukuma wiki). Results are recorded and compared across groups. Class discussion: why does nyama choma turn violet with Biuret reagent? How does this link to pepsin action in the stomach? Teacher uses wait time (minimum 15 seconds) before cold-calling 4 learners. Safety precautions for handling NaOH (Biuret) and ethanol are explicitly observed and discussed. Lesson 6 — EXPLAIN | Adaptations of the Human Digestive System: Learners examine labelled diagrams and photomicrographs (or digital images) of villus structure, stomach wall rugae, and the cardiac/pyloric sphincters. Teacher facilitates structured discussion: "What structural feature of the small intestine increases surface area? Why does the stomach not digest itself?" Learners construct an annotated diagram linking each structural adaptation to its function. Groups share one adaptation each in a gallery walk. DQB revisited — which original questions can now be answered with evidence? Lesson 7 — MODEL BUILDING | Making a Model of the Human Digestive System: Learners use locally available materials (cardboard tubes for oesophagus/intestines, a plastic bag for the stomach, cloth/sponge for villi, string for sphincters) to construct a physical 3D model of the alimentary canal. Each region must be labelled with (a) the organ name, (b) one mechanical and/or chemical process that occurs there, and (c) the enzyme(s) involved. Models are peer-assessed using a teacher-prepared checklist. Teacher circulates, asking probing questions: "Show me where the bile enters — what does bile do and why is that important for fat digestion?" Lesson 8 — DQB SYNTHESIS & ASSESSMENT | Driving Question Resolution and Unit Wrap-Up: Learners return to their Lesson 1 journey maps and original DQB predictions, annotating them in a different colour to show what they now know and what has changed. Class collaboratively answers the driving question in writing: "How is the human digestive system adapted to break down and absorb the nutrients in a typical Kenyan meal?" Teacher administers a structured food-test practical assessment (Benedict's test procedure, steps 1–4) and a short written component on enzyme-region matching. Unit closes with learners identifying one career connected to the sub-strand and explaining the connection.
Supporting Phenomena	– A ripe banana tastes sweet immediately, but raw unripe maize (corn on the cob) is starchy and not sweet — learners observe that both contain carbohydrates yet taste completely different, prompting questions about how digestive enzymes act on

	different forms of starch and sugar. – A learner who drinks a glass of whole milk in the morning feels full for longer than one who drinks a sugary soda — this observable difference in satiety prompts inquiry into how proteins and fats are digested more slowly than simple sugars, linking to the roles of the stomach and small intestine. – A patient at Kenyatta National Hospital who has had part of their small intestine removed must eat smaller, more frequent meals — this clinical observation (shared via a news clip or resource person) raises questions about why the small intestine is so critical for absorption and what specific adaptations (villi, microvilli) make it irreplaceable. – Mandazi (fried dough) left in the mouth for 30 seconds begins to taste slightly sweet — learners directly experience salivary amylase breaking down starch to maltose in real time, making enzyme action in the mouth a tangible, self-observed phenomenon.

LESSON 1 (40 minutes): The Ugali and Sukuma Wiki Puzzle — Launching the Anchoring Phenomenon	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	To launch the anchoring phenomenon by confronting learners with the genuine puzzle of how a familiar Kenyan meal is broken down and absorbed, activating prior knowledge and generating the unit's driving questions.
Knowledge	– Describe the visible components (starch, fibre, protein) present in a standard Kenyan meal of ugali, sukuma wiki, and nyama choma. – Recall that the human body extracts energy and nutrients from food through a process called digestion. – Identify, from prior knowledge, the major regions of the digestive system (mouth, stomach, small intestine, large intestine) and associate each with a general function.
Skills	– Construct an initial 'food journey map' by applying prior knowledge to sketch the path of food through the body. – Formulate testable questions and predictions about digestion by observing a real meal and identifying what is puzzling or unexplained. – Organise observations and questions onto a Driving Question Board (DQB) using the Notice–Wonder–Think framework.
Attitudes	– Appreciate that everyday Kenyan foods are scientifically complex systems worth rigorous investigation. – Show intellectual humility by acknowledging the limits of current understanding and welcoming uncertainty as the starting point of scientific inquiry.
Key Inquiry Question	How is the human digestive system adapted to its functions?
Purpose in Storyline	This lesson opens the entire unit storyline by presenting the anchoring phenomenon — the Ugali and Sukuma Wiki Puzzle — and establishing the three core mysteries (how the body knows which chemical to release, why digestion time differs for carbohydrates vs proteins, and how the body avoids digesting itself) that every subsequent lesson will work to resolve. Learners commit their initial mental models to paper before any instruction, giving the teacher a baseline and giving learners a record to revise as understanding grows.
Safety Notes	If real food is displayed, ensure no learner has a known food allergy (particularly to maize or meat proteins). Food items are for observation only — learners should not eat during the lesson. If photographs are used instead of real food, no specific hazards apply. Remind learners to wash hands before and after handling any food specimens.

B. LESSON OVERVIEW

This opening lesson exists to create genuine cognitive dissonance. The teacher places — or displays a high-resolution photograph of — a standard Kenyan school lunch: a serving of ugali (stiff maize-meal porridge), a portion of sukuma wiki (sautéed kale), and several pieces of nyama choma (grilled beef). Learners are immediately confronted with a sensory reality they have experienced hundreds of times but have never scientifically interrogated: three visibly different types of matter enter the mouth as solid, complex food, yet within hours the body has silently extracted energy, building materials, and regulatory molecules from all of them, expelling only waste. The puzzle is sharpened by three specific anomalies the teacher voices aloud — (1) the body appears to "know" which digestive chemical to deploy at each station along the alimentary canal, (2) the ugali is digested far faster than the nyama choma, and (3) the stomach, itself made of protein, somehow avoids being dissolved by its own acid and enzymes. None of these observations can be explained by the prior knowledge most Grade 10 learners bring, and that gap is the engine of the unit.

The lesson is structured around the PREDICT phase of the ARES framework. Before any content is taught, learners write individual predictions about the fate of each food component after swallowing, then sketch a first-draft 'food journey map' from memory. This activates and externalises prior knowledge, giving the teacher an immediate formative baseline and giving learners a personal stake in the investigation — they will return to these sketches at the end of the unit to measure their own growth. The lesson closes with the class co-constructing the unit's Driving Question Board (DQB), sorting their wonders into categories that map directly onto the lessons ahead.

This lesson deliberately does not resolve any of the three anomalies. Its purpose is to open inquiry, not close it. By the end of 40 minutes learners should feel productively puzzled — curious rather than confused — and should have committed at least one prediction to paper that they are not sure about. The cultural familiarity of ugali, sukuma wiki, and nyama choma is pedagogically essential: it ensures every learner, regardless of socio-economic background, has personal experiential data to draw on, and it grounds the entire unit's science in Kenyan lived reality.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Cellular respiration Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Digestion (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives a blank 'Prediction Sheet' with three columns labelled UGALI, SUKUMA WIKI, and NYAMA CHOMA and a fourth column labelled WHAT I AM NOT SURE ABOUT. Working individually and in silence (5 minutes), learners write at least one prediction per food item about what they think happens to that food after it is swallowed — using their own words, diagrams, or a mixture of both. Learners then share their predictions with a shoulder partner for 2 minutes before a whole-class cold-call share-out. The prediction sheet is kept by the learner and will be revisited at the end of the unit.	Before distributing the sheet, the teacher holds up (or projects) the food display and says: 'Look carefully at this plate. This is the same lunch many of you ate this week. I want you to think about what happens to this food after you swallow it — not what you have been told, but what YOU actually think happens. You have five minutes. Write or draw. There are no wrong answers right now.' Set a visible timer for 5 minutes. Circulate silently — do NOT correct or affirm any prediction. After 5 minutes, say: 'Turn to the person next to you. Share ONE prediction you are confident about and ONE thing you wrote that you are honestly not sure about. You have 2 minutes.' After partner talk, cold-call exactly 6 learners (aim for 2 high-confidence predictors, 2 uncertain predictors, 1 learner who drew a diagram, 1 learner who focused on the nyama choma). Use the prompt: 'Tell us one thing you predicted — and tell us WHY you predicted that.' Apply a minimum WAIT TIME of 12 seconds after each cold-call before accepting a response. Record key prediction phrases verbatim on the	Elicit-Before-Explain (EBE): By requiring predictions before any instruction, the teacher surfaces the full range of prior conceptions in the room — including likely misconceptions such as 'the stomach melts everything' or 'the body just absorbs whatever it needs.' Recording these verbatim on the board makes prior knowledge visible and creates a public record the class will audit and revise across the unit. This strategy is foundational to NGSS Science and Engineering Practice 1 (Asking Questions) and Practice 6 (Constructing Explanations).	The teacher scans Prediction Sheets during circulation to identify: (a) learners who can name specific organs (mouth, stomach, small intestine) — these learners have usable prior knowledge; (b) learners who describe only physical change ('the food gets mushed up') with no chemical language — these learners need the chemical digestion thread flagged early; (c) learners who leave the sukuma wiki or nyama choma column blank — these learners may have a carbohydrate-centric mental model that will need deliberate challenge. The teacher notes at least 3 specific misconceptions heard during partner talk for use in the Explain phase.

		board in three columns matching the food items, using learners' exact words without editing.		
Observe Phase 📹 **VIDEO:** Cellular respiration Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Digestion (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher directs learners' attention to the displayed meal (real or photographed) and leads a structured 'Slow Look' observation protocol. Learners spend 3 minutes observing the meal in silence and recording sensory observations on a 'Notice' card — colour, texture, visible structure, smell (if real food is present), and any changes they can already see happening to the food (e.g., the ugali drying, the sukuma wiki wilting). The teacher then poses three 'anomaly prompts' — three puzzling observations about digestion — which learners record on the reverse of their Notice card as 'Things I cannot yet explain.' Learners do not attempt to explain these anomalies yet; they simply record them as open questions.	Distribute index cards labelled NOTICE (front) and CANNOT YET EXPLAIN (back). Say: 'Before we do any science today, I want you to be a careful observer. Look at this meal for three minutes. On the NOTICE side, write everything you can observe — what you see, what you smell if the food is present, what you notice about the different textures. Do not explain anything yet. Just observe.' Set timer for 3 minutes. After the timer, say: 'Now I am going to share three puzzles with you. These are things that real scientists — including Kenyan nutritionists and medical researchers at the University of Nairobi and Kenyatta National Hospital — have spent careers studying. I want you to write each puzzle on the CANNOT YET EXPLAIN side of your card.' Read each anomaly slowly and clearly, pausing for 15 seconds after each one: PUZZLE 1 — 'Your body releases different digestive chemicals at different points along the alimentary canal. How does it know which chemical to release, and when?' PUZZLE 2 — 'The ugali in this meal is digested and absorbed much faster than the nyama choma. Why does starch digest faster than protein?' PUZZLE 3 — 'Your stomach is made of protein. Your stomach also produces acid and protein-digesting enzymes. So why does your stomach not digest itself?' After reading all three, say: 'These three puzzles are what we are going to spend this entire unit solving together.' Cold-call 3	Anomaly-Driven Noticing (a variant of NGSS Practice 1 — Asking Questions and Defining Problems): By presenting three specific, counter-intuitive anomalies tied directly to the familiar meal, the teacher transforms a routine food observation into a scientific puzzle. The 'Slow Look' protocol slows down perception and trains learners to distinguish observation from explanation — a critical scientific habit. Recording anomalies as 'things I cannot yet explain' rather than 'questions I have' positions learners as people who already have relevant observations, not as empty vessels waiting to be filled.	The teacher collects NOTICE cards at the end of the phase and quickly scans for: (a) learners who conflate observation with explanation on the NOTICE side (e.g., writing 'the ugali has carbohydrates' rather than 'the ugali is white and smooth and dense') — this reveals a tendency to skip observation and jump to conclusion, which the teacher will address in Lesson 3; (b) learners who show genuine surprise at Puzzle 3 (the stomach-digesting-itself anomaly) — these learners are intellectually engaged with the chemical digestion thread; (c) learners who draw structural diagrams of the food items — these learners may benefit from leading the model-building phase.

		learners and ask: 'Which of these three puzzles surprises you the most — and why?' Apply 12-second WAIT TIME before accepting each response.		
Explain Phase 📹 **VIDEO:** Cellular respiration Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Digestion (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The teacher facilitates a whole-class 'What Do We Already Know?' gallery sort. Prediction phrases collected from the Predict Phase (already written on the board) are now sorted by the class into three categories: CONFIDENT WE KNOW THIS, THINK WE KNOW BUT NOT SURE, and HAVE NO IDEA. Learners stand up, walk to the board, and physically move sticky-note labels (or the teacher moves them based on learner direction) to sort the prior-knowledge claims. The class then identifies at least one piece of prior knowledge per food item that they are confident is true and will serve as an anchor — this becomes the 'Known Foundation' that the DQB will build upon.	Transcribe 8–10 of the most interesting/most common prediction phrases from the board onto individual sticky notes (prepared during the Observe Phase while learners are writing). Draw three columns on a fresh section of the board: CONFIDENT | THINK SO | NO IDEA. Say: 'I have pulled out some of the things you predicted. As a class, let us sort them. I will read each one — if you think we KNOW this is true, show me a thumbs-up. If you THINK it might be true but you are not certain, show me an open hand. If you honestly have NO IDEA, show me a thumbs-down.' Read each sticky note, tally the hand signals aloud ('I see 18 thumbs-up, 7 open hands, 3 thumbs-down'), and place the sticky note in the majority column. After sorting, say: 'Look at our CONFIDENT column. These are the building blocks we already have. Now look at our NO IDEA column — that is our work for this unit.' Cold-call 2 learners from the NO IDEA column and ask: 'Can you say in one sentence what specifically you do not yet understand about this?' Apply 15-second WAIT TIME. Write their responses verbatim — these will seed the DQB. Do NOT answer any question at this stage. Say explicitly: 'I am not going to answer these today. That would spoil the investigation. But I am going to promise you this — by the end of this unit, YOU will be able to	Collaborative Knowledge Audit (linked to NGSS Practice 7 — Engaging in Argument from Evidence): The gallery sort externalises the epistemological status of prior knowledge claims — distinguishing confident knowledge from hunches from ignorance. This metacognitive move is crucial in CBE because it develops 'learning to learn' competency. By making uncertainty visible and public, the teacher normalises intellectual humility and frames the unit as a genuine investigation rather than a transmission of known facts. The sorting process also reveals consensus vs. disagreement within the class, which the teacher will exploit in Lessons 3–5 to create productive intellectual conflict.	The teacher notes which prior-knowledge claims land in the CONFIDENT column — these will be used as scaffolds in Lessons 2 and 3. The teacher specifically watches for: (a) whether learners can correctly identify that ugali is starch-based and nyama choma is protein-based — this is the chemical foundation of Lessons 4 and 5; (b) whether any learner mentions 'enzymes' — if so, the teacher writes the word on the board with a star and says 'Remember that word — we will need it'; (c) whether learners show confidence about mechanical digestion (chewing) but uncertainty about chemical digestion — this gap is the primary conceptual target of Lessons 2–4.

		answer every single one of these questions yourself.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Cellular respiration Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Digestion (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner receives three sticky notes in different colours: YELLOW = something I noticed about the meal, BLUE = something I wonder/a question I have, GREEN = something I think I already know. Learners individually write one idea per sticky note (3 minutes), then post their notes on the class DQB — a large sheet of paper or section of whiteboard divided into the three colour zones. The teacher facilitates a 5-minute whole-class clustering activity in which learners identify recurring themes and group similar questions together. The class agrees on the unit's Driving Question (pre-written by the teacher, now revealed): 'How is the human digestive system adapted to break down and absorb the nutrients in a typical Kenyan meal — and what happens when those adaptations fail?' Learners copy this into their science notebooks.	Distribute three sticky notes per learner. Say: 'You now have three minutes to capture your thinking on these sticky notes. YELLOW — one thing you noticed about the meal that you found interesting or surprising. BLUE — one question you genuinely have about what happens to this food inside your body. GREEN — one thing you are fairly sure you already know about digestion. One idea per note. Go.' Set timer for 3 minutes. After posting, stand back and say: 'Look at our board. What patterns do you see? Are there questions that keep appearing?' Cold-call 3 learners to identify clusters. Say: 'Let me group these — I can see questions about WHERE digestion happens, questions about HOW chemicals are involved, and questions about WHAT goes wrong. Does that match what you see?' After clustering, reveal the Driving Question on a pre-prepared card or slide. Say: 'This is the question that every lesson in this unit is designed to answer. Some of you have already written versions of this question on your blue notes — which means you are already thinking like scientists.' Point to 2–3 blue notes that are close paraphrases of the driving question. Say: 'We will come back to this board every lesson and ask ourselves: what new understanding can we add today?' Instruct learners: 'Write the Driving Question at the top of a fresh page in your notebook. Below it, write your three personal questions from your blue notes. These are YOUR investigation agenda for this unit.'	Notice–Wonder–Know (NWK) DQB Protocol (NGSS Practice 1 — Asking Questions; Practice 8 — Obtaining, Evaluating, and Communicating Information): The three-colour DQB is a living document that will be updated every lesson. By distinguishing NOTICE (empirical observation) from WONDER (question/hypothesis) from KNOW (prior knowledge claim), the board models the epistemological structure of science itself. Clustering questions into themes previews the unit's conceptual arc and gives learners agency in the investigation — their questions, not the teacher's, are visibly driving the unit forward.	The teacher photographs the completed DQB at the end of the lesson (for the ARES platform record) and scans for: (a) the sophistication of BLUE questions — are learners asking 'what happens to the food?' (surface) or 'why does the stomach not digest itself?' (deep) — this calibrates the conceptual level for Lesson 2; (b) whether any GREEN notes contain factual errors (e.g., 'the liver produces stomach acid') — these are flagged quietly and addressed in Lesson 3 without embarrassing the author; (c) whether the driving question resonates — if more than half the class has written a version of it on their blue notes, the phenomenon launch has succeeded.

Model Building Phase 📹 **VIDEO:** Cellular respiration Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Digestion (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners spend the final 6 minutes of the lesson sketching their first 'Food Journey Map' — a personal diagram showing what they currently believe happens to the ugali, sukuma wiki, and nyama choma from the moment they enter the mouth to the moment waste leaves the body. The map must include: (1) at least three labelled body regions, (2) at least one arrow showing direction of movement, and (3) at least one annotation explaining what the learner thinks is happening at each region. Learners write the date and the label 'DRAFT 1 — Before Investigation' on the map. The map is stored in the learner's science notebook and will be revised at the end of Lessons 3, 5, and 8.	Say: 'For the last few minutes, I want you to draw your current model of digestion — not what you have been taught, but what YOU currently believe happens. Draw the body — it can be a simple outline — and show me what you think happens to the ugali, the sukuma wiki, and the nyama choma as they travel through. Label at least three regions and write a note at each one explaining what you think is happening there. This is DRAFT 1. It does not need to be correct. It needs to be honest.' Set timer for 6 minutes. Circulate and observe — do NOT correct any diagram. If a learner is stuck, prompt: 'Where does the food go first after you swallow? Draw that.' After 6 minutes, say: 'Write on your map: DRAFT 1 — Before Investigation, and today's date. Do not erase anything. Every mistake on this page is data — it tells us where your understanding will grow the most.' Hold up two anonymised exemplars (use maps drawn on a previous run of this lesson, or teacher-prepared examples showing a simple and a more detailed attempt) and say: 'Both of these are valuable. Both will look very different by the end of this unit. I will ask you to compare your Draft 1 to your final model — and the bigger the difference, the more you will have learned.' Collect maps digitally (photograph with ARES platform camera) or keep in notebooks.	Initial Model Construction (NGSS Practice 2 — Developing and Using Models): The 'Food Journey Map' is the first iteration of the cumulative model that learners will revise throughout the unit. Research on model-based learning shows that committing an initial model to paper — however incomplete — dramatically increases engagement with subsequent evidence that challenges or refines the model. By labelling it 'DRAFT 1' and explicitly honouring incorrect models as 'data,' the teacher establishes a classroom norm of revision rather than replacement, which is central to scientific thinking and to the CBE competency of 'learning to learn.'	The teacher uses the maps as the primary diagnostic for Lesson 2 planning. Specifically, the teacher looks for: (a) whether learners include the small intestine as a distinct region — if fewer than half the class do so, Lesson 2 must open with a physical model of the alimentary canal to make the small intestine's length and structure concrete; (b) whether any learner draws enzymes or chemicals (even labelled vaguely as 'digestive juice') — these learners are ready for the enzyme specificity thread in Lesson 4; (c) whether learners treat ugali, sukuma wiki, and nyama choma as having different fates, or whether they draw all three food items taking the same path — if the latter, the mechanical vs. chemical digestion distinction in Lesson 2 needs to be anchored explicitly to the three foods from the phenomenon.

D. TEACHER REFLECTION

1. When learners sorted prior-knowledge claims into CONFIDENT / THINK SO / NO IDEA, which category was most populated — and does the distribution match what you expected from a Grade 10 class entering this sub-strand? What does this tell you about where to spend the most instructional time in Lessons 2–4?

2. Look at the BLUE wonder-questions on the DQB. Did any learner's question go significantly beyond the three anomalies you posed — for example, asking about digestion in relation to nutrition-related disease (diabetes, malnutrition, ulcers)? If so, how will you honour that question within the unit storyline? 3. Review the Draft 1 Food Journey Maps. How many learners drew the small intestine as a distinct region? How many drew the large intestine? What does the absence of a specific organ tell you about where prior teaching at Junior School may have created gaps? 4. Did the cultural anchor — the ugali, sukuma wiki, and nyama choma meal — generate genuine engagement across ALL learners in the room, including those from non-Kenyan culinary backgrounds or those with dietary restrictions? If any learner appeared disengaged or excluded, how will you adjust the phenomenon framing in Lesson 2? 5. At the moment you revealed the Driving Question, how many learners visibly recognised it as a version of their own blue-note question? If fewer than half did, the anomaly prompts in the Observe Phase may need to be sharpened — consider whether Puzzles 1–3 need to be reworded to more directly echo the language your specific learners used in their predictions.

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed a standard Kenyan school lunch (ugali, sukuma wiki, nyama choma) and noticed that three visibly different types of matter — starchy, fibrous, and protein-rich — enter the digestive system yet are all processed by the same body. They recorded specific anomalies: the body's apparent 'knowledge' of which digestive chemical to deploy, the different digestion times for starch versus protein, and the mystery of why the stomach does not digest itself.
What did I learn?	Learners surfaced and organised their prior knowledge about digestion, distinguishing confident knowledge from uncertain hunches and genuine ignorance. They identified the major regions of the digestive system from memory, recognised that ugali is primarily starch, sukuma wiki is primarily fibre and vitamins, and nyama choma is primarily protein, and formulated at least one personal investigation question that they will pursue across the unit.
How does this explain the phenomenon?	This lesson does not yet explain the phenomenon — it deliberately opens the puzzle. Learners' Draft 1 Food Journey Maps represent their current incomplete model of digestion. The three anomalies posed by the teacher have created cognitive dissonance that makes the rest of the unit's investigation necessary and motivating. The Driving Question Board now anchors all future learning to the central question: how is the human digestive system adapted to break down and absorb the nutrients in a typical Kenyan meal — and what happens when those adaptations fail?

LESSON 2 (40 minutes): Mechanical vs. Chemical Digestion — Breaking Down the Ugali and Sukuma Wiki	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners gather direct observational evidence of how the mouth and stomach begin breaking down the complex substances in ugali, sukuma wiki, and nyama choma through two fundamentally different types of digestion.
Knowledge	– Distinguish between mechanical digestion (physical breakdown of food by teeth, tongue, and stomach churning) and chemical digestion (breakdown of food molecules by enzymes and acid). – Identify the roles of teeth, tongue, and salivary glands in the mouth, and the roles of pepsin and hydrochloric acid (HCl) in the stomach. – Explain why the starch in ugali begins to taste sweet after chewing, as evidence that salivary amylase acts on carbohydrates in the mouth.
Skills	– Observe and record evidence from an animation/simulation and a live mandazi demonstration, using a structured observation table. – Construct a two-column comparison chart distinguishing mechanical from chemical digestion across the mouth and stomach.
Attitudes	– Appreciate that the human body operates as a precisely regulated chemical system, with each organ releasing the correct substance at the correct time. – Show curiosity and openness when familiar experiences (chewing ugali) are re-examined through a scientific lens.
Key Inquiry Question	How is the human digestive system adapted to its functions?
Purpose in Storyline	Lesson 1 surfaced what learners already believe about digestion. Lesson 2 provides the first concrete observational evidence — anchored in the mouth and stomach — that the body uses two distinct strategies (mechanical and chemical) to begin dismantling the ugali, sukuma wiki, and nyama choma. This resolves part of the puzzle (why does the food change form?) while generating new, deeper questions (who controls which enzyme is released and when?) that will drive the remaining lessons.
Safety Notes	Learners should not share food items due to hygiene concerns. If the mandazi demonstration involves learners chewing, each learner must use their own piece. Wash hands before handling any food. No specific chemical hazards in this lesson.

B. LESSON OVERVIEW

This lesson is the first evidence-gathering session in the unit and marks the transition from the predictions and prior-knowledge surfacing of Lesson 1 into direct observation. The anchoring phenomenon — a school lunch of ugali, sukuma wiki, and nyama choma — is kept physically visible at the front of the room throughout. Learners begin by predicting what they think happens to a piece of ugali the moment it enters the mouth, then gather layered evidence: first from a short animation/simulation of mechanical and chemical digestion along the alimentary canal, and then from a live teacher demonstration called the "Mandazi Moment" in which learners chew a small piece of mandazi (a familiar, starchy Kenyan snack similar in composition to ugali) for 60 seconds and observe the gradual sweetening taste — direct sensory evidence of salivary amylase converting starch to maltose in real time.

The lesson's central intellectual move is the construction of a two-column comparison chart (Mechanical vs. Chemical Digestion) that learners build incrementally as evidence accumulates. This chart is not handed to learners pre-made; instead it grows from their own observations and is explicitly connected back to the ugali-sukuma wiki-nyama choma meal at every step. By the end of the Explain Phase, learners can account for why the starchy ugali begins to break down in the mouth while the protein in nyama choma does not — because the enzyme present in saliva (amylase) targets starch, not protein. This partial explanation is deliberately incomplete:

learners will notice that protein digestion in the stomach raises the unsettling question of why the stomach does not digest its own walls — a question flagged on the Driving Question Board (DQB) for future investigation.

The Model Building Phase asks learners to revise the simple 'food travels through a tube' sketch most drew in Lesson 1 into a labelled diagram that now includes the mouth (with teeth, tongue, salivary glands) and stomach (with stomach wall muscles, pepsin, HCl), with arrows indicating where mechanical and chemical digestion occur. This cumulative model revision is the core sensemaking artefact of the lesson and will be carried forward and revised in every subsequent lesson of the unit.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Cellular respiration Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Mechanical Weathering (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners look at the ugali and sukuma wiki meal displayed at the front of the room (or a photograph on screen). In their science notebooks, each learner individually draws a quick sketch of the inside of the mouth and writes a 2-sentence prediction answering: 'What do you think happens to this lump of ugali in the first 60 seconds after it enters your mouth — and is anything chemically different about it by the time you swallow?' Learners then share predictions with a seat-partner before three pairs are cold-called to share with the class.	Teacher re-displays the meal from Lesson 1 and says: 'Last lesson we asked big questions about this meal. Today we start gathering evidence. Before we observe anything, I want your best prediction — right or wrong, it doesn't matter yet.' Allow 3 minutes of silent individual writing. After 3 minutes, say: 'Turn to your partner. You have 90 seconds. Share your prediction and listen to theirs.' After partner talk, cold-call exactly 3 pairs (select one high-confidence, one uncertain, one who drew something unexpected). For each, ask: 'What made you predict that?' [WAIT TIME: 12 seconds after each follow-up question before accepting a response.] Record key prediction words on the board in a column labelled 'We Predict' — do not correct errors at this stage. Close with: 'Hold your prediction in your mind. We are about to test it.'	Predict-Before-Observe (PBO): By committing to a prediction in writing before any instruction, learners activate prior schema and create a personal 'expectation gap' that makes subsequent evidence more cognitively salient. Displaying predictions publicly on the board also makes misconceptions visible to the teacher and creates a reference point for revision later in the lesson.	Teacher scans written predictions as learners write (circulate for 2 minutes). Look for: (1) learners who conflate mechanical and chemical digestion (e.g., 'teeth break the food and that is all that happens'); (2) learners who spontaneously mention saliva doing something chemical; (3) learners who mention taste changing. These three groups represent different readiness levels. Cold-call responses reveal whether learners distinguish physical change from chemical change — a key conceptual threshold for this lesson.
Observe Phase 📹 **VIDEO:** Cellular respiration Source: Khan Academy (English - US curriculum) 🔍 Search ARES	Learners watch a 5–6 minute annotated animation/simulation of food moving from the mouth through the oesophagus to the stomach, pausing at key moments to complete a structured Observation Table in their notebooks (columns: Location |	Before the animation: 'Open your Observation Table. Your job is to be a scientist, not a passive watcher. Every time you see food change — physically or chemically — write it down. Pause your pencil when you need to look, then write.' Play the animation (pause at the	Multi-Modal Evidence Layering: Learners gather evidence through three distinct sensory channels — visual (animation), proprioceptive/gustatory (mandazi chewing), and diagrammatic (labelled structures). Each channel reinforces and extends the others,	Circulate during mandazi chewing and observe whether learners are writing at the 20/40/60-second intervals — a proxy for engagement with the task. After cold-calling 4 learners for taste-change words, listen for: sweetness, softer, watery, no

for similar videos 📖 **HTML:** Mechanical Weathering (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	What I see happening physically | What I see happening chemically | My question). Immediately after the animation, the teacher conducts the live 'Mandazi Moment' demonstration: each learner receives one small, plain piece of mandazi and is instructed to chew it slowly for 60 seconds without swallowing, paying attention to any changes in taste. Learners record their sensory observations in the table. The class then briefly views a still image labelling the mouth structures (incisors, molars, tongue, salivary glands) and a diagram of the stomach wall with labels for gastric glands, pepsin, and HCl.	mouth segment after ~90 seconds): 'What did you just observe? Write it — don't talk yet.' [WAIT TIME: 15 seconds.] Resume animation. Pause again at stomach churning: 'What is the stomach doing that teeth were also doing? Write one word.' [WAIT TIME: 10 seconds.] After animation, distribute mandazi pieces: 'This is science, not a snack break. Chew slowly. Count to 60. Notice your tongue. Does the taste change? Write exactly what you notice at 20 seconds, 40 seconds, 60 seconds.' After 60 seconds, cold-call 4 learners: 'Describe the taste change in one word.' [WAIT TIME: 12 seconds each.] Display mouth and stomach diagrams: 'Use these diagrams to add labels to your Observation Table. Match what you observed to a structure.'	reducing the likelihood that any single mode is misunderstood. The Observation Table forces learners to sort evidence into physical vs. chemical categories in real time, building the conceptual distinction that is the lesson's core knowledge target.	change. 'No change' responses indicate the learner may not have chewed long enough or may have a conceptual block about chemical change being perceptible — note these learners for targeted questioning in the Explain Phase. Check Observation Tables for completeness of the physical vs. chemical columns.
Explain Phase 📹 **VIDEO:** Cellular respiration Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Mechanical Weathering (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Using their Observation Tables as raw data, learners collaboratively construct a two-column comparison chart: 'Mechanical Digestion' vs. 'Chemical Digestion.' Working in groups of four, they sort their observations into the two columns, then add the agent responsible (e.g., teeth, stomach muscles for mechanical; salivary amylase, pepsin, HCl for chemical) and the food component acted upon (starch in ugali, protein in nyama choma, fibre in sukuma wiki). Each group then writes a single-sentence answer to: 'Why did the ugali taste sweet after 60 seconds of chewing but the nyama choma would not?' Groups share answers; the class reaches a consensus explanation. Teacher then introduces the concept of enzyme specificity using a 'lock-	Distribute blank two-column chart templates (or have learners draw them). Say: 'Use only your Observation Table. Sort every observation into mechanical or chemical. If you are not sure, put it in the middle.' [Allow 4 minutes group work.] Circulate and listen for groups who place 'saliva' under mechanical digestion — this is a common error. Prompt: 'Saliva is a liquid. Can a liquid physically crush food?' [WAIT TIME: 12 seconds.] After groups share charts, cold-call one group per column to justify one entry: 'Why did you put stomach churning under mechanical? What evidence from the animation supports that?' [WAIT TIME: 15 seconds.] Pose the key explanatory question to the whole class: 'Why sweet after chewing ugali but not after	Evidence-to-Explanation Reasoning (Claim-Evidence-Reasoning lite): Learners are not told the distinction between mechanical and chemical digestion — they construct it from their own sorted observations. The two-column chart is an external representation that makes reasoning visible. The 'lock-and-key' analogy contextualised with a jembe bridges the abstract concept of enzyme specificity to a concrete Kenyan agricultural tool learners know intimately, reducing cognitive load while preserving scientific accuracy.	Collect one two-column chart per group at the end of the phase (or photograph with a phone/tablet if available). Evaluate: (1) Are mechanical and chemical correctly sorted? (2) Is salivary amylase correctly identified as acting on starch (ugali) not protein (nyama choma)? (3) Does the group's single-sentence explanation include the word 'enzyme' or 'amylase'? Groups that cannot distinguish the two types of digestion or who cannot connect amylase to ugali starch need re-teaching before Lesson 3. The self-acid digestion question is a stretch indicator — any learner who raises it spontaneously (rather than in response to teacher prompt) is demonstrating advanced inferential reasoning.

	and-key' analogy tied to the Kenyan context: 'Salivary amylase is like a jembe (hoe) designed only to break up maize starch — it cannot break up the protein fibres in nyama choma. The stomach has a different tool — pepsin — but pepsin only works when HCl makes the stomach very acidic.'	chewing nyama choma?' Cold-call 3 learners before offering the lock-and-key analogy. After the analogy, ask: 'So the stomach makes HCl — a strong acid. If pepsin dissolves protein… what stops the stomach dissolving itself?' [WAIT TIME: 15 seconds.] Do NOT answer this question — flag it explicitly for the DQB.		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Cellular respiration Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Mechanical Weathering (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Each learner writes one new question that arose during today's lesson on a sticky note (or a slip of paper) using the stem: 'I now wonder…' or 'This lesson made me ask…' Learners post their questions on the class DQB (a section of the board or a dedicated chart paper). The teacher reads all questions aloud and, together with learners, clusters them into three emerging themes: (A) How the body controls which enzyme is released and when; (B) How the stomach avoids digesting itself; (C) What happens to sukuma wiki fibre and nyama choma protein after the stomach. Learners also revisit the DQB entries from Lesson 1 and tick any that today's evidence has partially answered.	Distribute sticky notes. Say: 'Today you observed something real happening inside your own mouth. That always raises new questions. Write the one question that is most alive for you right now — not a question you think I want, your actual question.' [Allow 2 minutes silent writing.] As learners post notes, read each one aloud in a neutral voice: 'Someone asks: why doesn't the stomach digest itself? That goes here — cluster B.' After clustering, say: 'Look at our DQB from Lesson 1. Which questions did today's evidence help us answer — even partially?' Cold-call 2 learners to identify a question and explain how today's observation addressed it. [WAIT TIME: 12 seconds each.] Close: 'Notice cluster B — the stomach question. We deliberately left it unanswered today. We will return to it. That is how science works: every answer opens a door to a harder question.'	Driving Question Board — Progressive Clustering: The DQB functions as a public, persistent record of the class's evolving understanding. Clustering questions into themes (rather than leaving them as an undifferentiated list) helps learners see the structure of the unit's intellectual territory. Ticking partially-answered questions from Lesson 1 makes the process of sensemaking visible and gives learners a sense of genuine scientific progress, sustaining motivation across the 8-lesson sequence.	Read all sticky-note questions before the next lesson. Questions that indicate a productive conceptual struggle (e.g., 'Why does HCl not burn the stomach?', 'Does sukuma wiki get digested differently because it has cellulose?') signal learners who are ready for extension. Questions that reveal persistent confusion about the basic mechanical/chemical distinction (e.g., 'What is the difference between chewing and digesting?') signal learners who need the Lesson 2 concepts reinforced at the start of Lesson 3. Count how many learners independently identified a Lesson-1 DQB question as partially answered — a proxy for metacognitive engagement.
Model Building Phase 📹 **VIDEO:** Cellular respiration Source: Khan Academy (English - US curriculum) 🔍 Search ARES	Learners retrieve the simple 'food tube' sketch they drew in Lesson 1. Using a different coloured pen, they now revise and extend it into a labelled diagram that includes: the mouth (labelled: incisors for cutting, molars for grinding, tongue for mixing, three pairs of salivary glands releasing amylase), the	Say: 'Take out your Lesson 1 model. This is your scientific model — it should grow every lesson as you learn more. Use a different colour so we can see what is new today.' Display the Lesson 1 model template on the board beside today's labelled diagrams. After 5 minutes of individual revision:	Cumulative Model Revision (Progressive Modelling): The model is not a one-off product but a living artefact revised at the end of every lesson. Using a different colour for each lesson's additions makes the growth of understanding literally visible. Connecting the model revision to	Collect models (or photograph them) at the end of the lesson. Evaluate against three criteria: (1) Are mouth structures (teeth types, salivary glands) correctly labelled? (2) Is the stomach shown with both mechanical [M] and chemical [C] labels? (3) Is there at least one arrow showing a food component

for similar videos 📖 **HTML:** Mechanical Weathering (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	oesophagus (labelled: peristalsis — mechanical), and the stomach (labelled: muscular walls for churning — mechanical; gastric glands releasing pepsin and HCl — chemical; showing acidic environment). On each label, learners write a bracketed note indicating whether that structure contributes to mechanical digestion [M], chemical digestion [C], or both [M+C]. Learners also draw an arrow from the ugali meal image to the mouth with the annotation: 'Starch → maltose (salivary amylase, mouth).'	'Hold up your model. Show me: where have you added something new in today's colour?' Scan the room for models that are unchanged from Lesson 1 — approach those learners quietly and ask: 'What did you observe today that was not in your Lesson 1 model?' Cold-call 2 learners to describe one revision they made and why: 'What evidence from the animation — or from your own mouth — made you add that label?' [WAIT TIME: 12 seconds.] Close: 'Your model is not finished. It is missing the small intestine, the large intestine, enzymes from the pancreas, and much more. We will keep building. Scientists never have a finished model — they only have a better one than last time.'	specific observational evidence (the animation, the mandazi chewing) reinforces that models must be grounded in evidence, not imagination — a core NGSS Science and Engineering Practice. The deliberate incompleteness of the model sustains intellectual tension across the unit.	(starch from ugali) being acted on by a named enzyme (amylase)? Models that show only physical structures without enzyme labels indicate that the chemical digestion concept has not yet been consolidated and must be revisited at the start of Lesson 3.

D. TEACHER REFLECTION

1. During the Mandazi Moment, how many learners reported a change in taste, and how many reported no change? For those who reported no change, did I probe their chewing behaviour, or did I move on too quickly — and how will I address this at the start of Lesson 3?

2. When I posed the question 'Why doesn't the stomach digest itself?', did learners engage with genuine intellectual discomfort, or did they deflect with surface answers? What does this tell me about whether the cognitive dissonance generated by the anchoring phenomenon is still alive for this class?

3. Looking at the two-column comparison charts collected at the end of the Explain Phase, which groups correctly distinguished mechanical from chemical digestion, and which placed 'saliva' or 'stomach acid' under the wrong column? What targeted question or analogy will I use at the start of Lesson 3 to address this specific error without re-teaching the whole lesson?

4. Reviewing the DQB sticky notes: are learners' questions becoming more specific and scientifically precise compared to Lesson 1 (e.g., moving from 'how does digestion work?' to 'why does amylase only work on starch?')? If not, what scaffolding do I need to provide for question formation?

5. Did the 'jembe' lock-and-key analogy for enzyme specificity land successfully across all learners, including those from non-farming households in Nairobi urban settings? Should I prepare an additional analogy (e.g., a specific key for a specific padlock) to ensure the concept is accessible to learners unfamiliar with agricultural tools?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Learners observed (via animation) food being physically crushed and mixed by teeth, tongue, and stomach muscles, and (via the Mandazi Moment — chewing their own piece of mandazi for 60 seconds) that the taste of a starchy food changes from bland to slightly sweet, providing direct sensory evidence that a chemical change occurs in the mouth.
What did I learn?	The body uses two distinct strategies to break down food: mechanical digestion (physical breakdown by teeth, tongue, and

	stomach churning — changes the size and shape of food but not its chemical identity) and chemical digestion (breakdown of food molecules by specific enzymes — salivary amylase in the mouth acts on starch, and pepsin with HCl in the stomach acts on protein). Each enzyme acts on a specific type of food molecule.
How does this explain the phenomenon?	This explains part of the ugali-sukuma wiki puzzle: the ugali (starch) begins to change chemically the moment it enters the mouth because salivary amylase is already present and specific to starch — which is why it tastes sweet after chewing. The nyama choma (protein) does not taste different in the mouth because amylase cannot act on protein. A new, deeper part of the puzzle is now exposed: if pepsin in the stomach dissolves protein, why doesn't the stomach dissolve its own protein-rich walls?

LESSON 3 (40 minutes): Enzymes and Regions of the Alimentary Canal — The Body's Chemical Toolkit	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners research and map the specific enzymes active in each region of the alimentary canal, discovering how the body deploys a precise chemical toolkit — region by region — to break down the ugali, sukuma wiki, and nyama choma in the anchoring phenomenon.
Knowledge	– Name the enzymes active in each region of the alimentary canal: salivary amylase (mouth), pepsin and rennin (stomach), pancreatic amylase, lipase, trypsin, and intestinal peptidases (small intestine). – State the substrate and product of each named enzyme. – Explain why different enzymes are secreted at different regions and why this sequential release is necessary for complete digestion.
Skills	– Retrieve, evaluate, and synthesise information from print and digital sources (Kenya Education Cloud / resource sheets) on enzyme activity by region. – Construct and present a region-by-region enzyme summary chart using digital tools or manila paper. – Communicate scientific findings clearly to peers through group presentation.
Attitudes	– Appreciate the elegant, coordinated design of the digestive system and how it solves the chemical challenge posed by a mixed Kenyan meal. – Demonstrate intellectual curiosity and persistence when researching unfamiliar enzyme names and their roles.
Key Inquiry Question	How is the human digestive system adapted to its functions?
Purpose in Storyline	Lessons 1 and 2 established that the body uses mechanical and chemical strategies to break down food, but left unanswered the mystery of HOW the body 'knows' which chemical to release at each stage. Lesson 3 directly attacks that mystery: learners discover that specific enzymes are deployed in a precise regional sequence along the alimentary canal, moving the class from 'the body does something chemical' to 'specific enzymes act on specific substrates in specific locations — and this is why the ugali's starch, the nyama choma's protein, and the sukuma wiki's fats are each handled differently.'
Safety Notes	No laboratory chemicals or hazardous materials are used in this lesson. Learners handle only printed resource sheets, digital devices, manila paper, and markers. Remind learners to handle devices responsibly and to return all printed materials in good condition for reuse.

B. LESSON OVERVIEW

This lesson is the third in an eight-lesson evidence-gathering sequence and sits at the heart of the OBSERVE phase of the unit storyline. Having spent Lesson 1 surfacing prior knowledge about digestion and Lesson 2 distinguishing mechanical from chemical digestion, learners are now ready to go deeper: they will investigate the specific enzymes operating in each anatomical region of the alimentary canal. The central cognitive challenge shifts from 'what types of digestion exist?' to 'which precise chemical agents act where, on which food molecules, and why?' This directly addresses the first puzzling question raised by the anchoring phenomenon — how does the body 'know' which chemical to release at each stage of digesting a Kenyan school lunch of ugali, sukuma wiki, and nyama choma?

Learners work in research groups (4–5 per group) using Kenya Education Cloud resources or printed resource sheets to investigate enzyme activity by region. Each group is assigned one or two regions of the alimentary canal and constructs a region-by-region enzyme summary chart, recording: the region, the enzyme(s) present, the gland or organ secreting the enzyme, the substrate acted upon, and the product produced. Groups then share findings in a brief gallery-walk and cold-called presentation, building a whole-class composite understanding of the full alimentary canal enzyme map. The teacher cold-calls exactly 3 groups to explain one enzyme each in detail.

This lesson is deliberately placed before the food-test experiments (Lesson 4) so that learners develop conceptual understanding of enzyme specificity before they generate their own empirical data. The enzyme chart produced today will serve as a reference anchor for all remaining lessons. The DQB is updated to capture newly answered questions and newly surfaced ones — particularly the question of how the body avoids digesting itself (the role of mucus and enzyme activation), which will be revisited in Lesson 5.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Digestive System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Human Digestive System (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Before any research begins, learners individually write predictions in their science notebooks in response to two questions displayed on the board: (1) 'At which stage of digestion do you think most chemical breakdown of ugali's starch happens — the mouth, stomach, or small intestine? Why?' and (2) 'Does your body use the same chemical to break down ugali (carbohydrate) as it uses to break down nyama choma (protein)? Explain your reasoning.' After 3 minutes of silent individual writing, learners share predictions with a partner (Think-Pair-Share), then 4 volunteers share with the class before the teacher collects a quick show-of-hands vote on the most popular predictions.	Display both prediction questions on the board or a chart. Say: 'Before we look at any resources today, I want your best scientific guess — not the right answer, your current thinking. Write in your notebooks for 3 minutes. Silence please.' [Wait 3 minutes — do not circulate or hint.] Then: 'Now turn to your partner and compare predictions. You have 90 seconds.' After pair talk, cold-call 4 learners: 'Amina, what did you and your partner predict for question 1?' [WAIT TIME: 10 seconds before prompting.] After each response, resist affirming or correcting — instead say: 'Interesting. Let's hold that prediction and see what the evidence tells us.' Conduct a show-of-hands: 'Hands up if you predicted the mouth is the main site of starch digestion… now hands for stomach… small intestine.' Record the class vote visibly on the board as a baseline to revisit after the Observe phase.	Prediction Baseline with Visible Vote Tally — by committing predictions to writing and recording the class vote publicly, learners create a cognitive anchor against which new evidence will be measured. This activates prior knowledge from Lessons 1 and 2, surfaces misconceptions (most learners predict the stomach as the primary site for all digestion), and creates genuine motivation to find out who was right. The visible tally on the board remains displayed throughout the lesson so learners can self-correct in real time.	Teacher scans written predictions during pair-share by circulating briefly. Look for: (1) learners who cannot distinguish between mechanical and chemical digestion (a gap from Lesson 2 — note these learners for targeted support during research); (2) learners who already mention specific enzymes by name (gifted students to challenge with 'enzyme activation' questions during research); (3) whether learners apply the phenomenon context (ugali, nyama choma) or revert to generic language. The show-of-hands vote gives a rapid whole-class formative snapshot — if fewer than 20% predict the small intestine as the primary site, note this as a key misconception to address explicitly in the Explain phase.
Observe Phase 📹 **VIDEO:** Digestive System Structure and Function - Overview	Learners work in pre-assigned research groups of 4–5. Each group receives a printed resource sheet (or accesses the Kenya Education Cloud on a shared device) covering enzyme activity in the alimentary canal. Groups are	Distribute resource sheets and/or direct groups to the Kenya Education Cloud. Say: 'Your group has been assigned a specific region of the alimentary canal — the region is written on your resource sheet. Your job is to	Jigsaw Expert Research with Gallery Walk Annotation — the jigsaw structure ensures every group develops genuine regional expertise rather than passively reading a full overview. The gallery walk transforms individual group	During circulation, teacher checks each group's chart for three specific accuracy markers: (1) correct substrate listed (starch for amylase, protein for pepsin/trypsin, fats for lipase — not generic 'food'); (2) correct secreting organ

Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Human Digestive System (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	assigned specific regions: Group 1 — Mouth; Group 2 — Stomach; Group 3 — Small Intestine (pancreatic enzymes); Group 4 — Small Intestine (intestinal enzymes / brush border); Groups 5+ repeat assignments if class is large. Each group reads their resource, discusses within the group, and constructs a region-by-region enzyme summary chart on manila paper or a digital slide. The chart must include five columns: Region | Enzyme Name | Secreted By | Substrate (Food Acted On) | Product. Groups have 12 minutes for research and chart construction. In the final 3 minutes of this phase, groups conduct a rapid gallery walk — posting their charts and circulating to read other groups' charts silently, adding sticky-note questions or annotations where they are uncertain.	become the class experts on that region. Construct your enzyme chart on the manila paper — every column must be filled. You have 12 minutes.' [Set a visible countdown timer.] Circulate actively during the 12 minutes. Stop at each group for approximately 90 seconds. At each stop, ask ONE probing question — do not answer: 'What does the enzyme actually DO to the ugali starch — what is it breaking the starch INTO?' or 'Which organ makes pepsin — and why does it make sense that the stomach makes this enzyme rather than the mouth?' or 'Look at the product column for lipase — can you connect this to what we said in Lesson 2 about fats?' After 12 minutes, announce the gallery walk: 'Post your charts on the wall. You have 3 minutes to silently read every other group's chart. Use a sticky note or pencil annotation to mark anything you find surprising or unclear.' During the gallery walk, teacher reads all charts quickly and mentally notes errors to address in the Explain phase — do not correct charts publicly yet.	products into a shared class resource, and the sticky-note annotations make uncertainty visible and discussable. This mirrors NGSS Science and Engineering Practice 8 (Obtaining, Evaluating, and Communicating Information) by requiring learners to retrieve information from a source, evaluate its relevance, and communicate it in a structured format to peers.	named (salivary glands, gastric glands, pancreas, intestinal glands); (3) a product listed that is chemically more specific than 'smaller pieces' (e.g., maltose for salivary amylase, peptides for pepsin, fatty acids and glycerol for lipase). Teacher mentally flags groups with errors in column 3 or 4 for targeted cold-calling in the Explain phase. Sticky-note questions on the gallery walk reveal the most common class-wide gaps — teacher collects these notes at the end of the gallery walk and reads them aloud in the Explain phase.
Explain Phase 📹 **VIDEO:** Digestive System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Human Digestive System (Flexbook) Source: CK-12	Groups return to their seats. The teacher leads a whole-class debrief structured around cold-called group presentations. Three groups are cold-called, each asked to explain ONE specific enzyme from their chart in detail, connecting it to the ugali-sukuma wiki-nyama choma phenomenon. After each cold-called explanation, the class is invited to add, correct, or question. The teacher then delivers a 5-minute consolidating explanation using the completed	Begin cold-calling: 'Group 2 — Stomach — Korir, please stand and explain to us what pepsin does to the nyama choma protein. Tell us: what is the substrate, what organ makes it, and what is the product.' [WAIT TIME: 12 seconds before any prompting.] After Korir's response, ask the class: 'Does anyone want to add to or challenge what Korir said? Hands.' Take 1–2 responses. Then: 'Group 1 — Mouth — Wanjiru, explain salivary amylase and connect it to	Cold-Called Jigsaw Debrief with Phenomenon Anchoring — by requiring cold-called groups to explain their specific enzyme in the context of the anchoring phenomenon (ugali, nyama choma), the debrief strategy prevents rote recitation and forces application. The teacher's use of targeted follow-up probes ('what is the product?', 'why the small intestine?') pushes learners to the level of mechanism, not just vocabulary. The 5-minute	Teacher listens for three quality indicators during cold-called responses: (1) correct use of substrate-product language (not just 'breaks it down' but 'breaks starch into maltose'); (2) ability to name the secreting organ correctly; (3) ability to connect to the phenomenon without prompting ('the ugali contains starch, so salivary amylase acts on it first'). If a cold-called learner cannot respond after full wait time, teacher asks the group: 'Can

🔍 Search ARES for similar readings	charts on the wall as reference, explicitly addressing: (1) why enzyme specificity matters — why salivary amylase cannot break down protein; (2) the sequential logic of enzyme deployment — why it is adaptive that protein digestion begins in the acid environment of the stomach rather than the mouth; (3) the introduction (first mention) of the concept that enzymes are specific to their substrates, connecting to the puzzle question of how the body avoids digesting itself (mucus lining — flagged for Lesson 5). Learners annotate their own charts with corrections and additions in a different coloured pen.	the ugali on our lunch tray.' [WAIT TIME: 10 seconds.] After response, probe: 'Wanjiru says salivary amylase breaks down starch. What is the product — what does the ugali starch become?' [WAIT TIME: 10 seconds — cold-call a different learner from Group 1 if Wanjiru cannot answer.] Third cold-call: 'Group 3 or 4 — Ochieng, explain ONE pancreatic enzyme of your choice and tell us why the small intestine needs so many different enzymes when the mouth and stomach only have one or two each.' [WAIT TIME: 15 seconds — this is a higher-order question; allow full wait time.] After the three cold-calls, deliver the 5-minute consolidating explanation. Use the wall charts as a visual map. Say: 'Now look at all four charts together — what do you notice about the journey from mouth to small intestine in terms of the number of enzymes?' Pause for 10 seconds of class observation before taking responses. Close by returning explicitly to the phenomenon: 'So when Waweru eats ugali at lunch, his salivary amylase starts working in his mouth — but the starch digestion is NOT finished there. Why not? Where does it continue?' [Cold-call one more learner — not previously called — for a closing response.]	consolidating explanation models expert scientific synthesis — demonstrating how individual pieces of evidence (each group's chart) combine into a coherent explanatory framework, mirroring NGSS Science and Engineering Practice 6 (Constructing Explanations).	someone from Group 2 help Korir?' — this reveals whether the gap is individual or group-wide. Teacher also collects the sticky-note annotations from the gallery walk and identifies the top 3 unanswered questions to post on the DQB. At close of Explain phase, teacher asks learners to give a silent thumbs-up (confident), thumbs-sideways (somewhat clear), or thumbs-down (still confused) — scan the room and note the proportion of sideways/down for reflection.
Driving Question Board (DQB) Creation 📹 **VIDEO:** Digestive System Structure and Function - Overview	Learners revisit the Driving Question Board established in Lesson 1 and updated in Lesson 2. Working individually for 2 minutes, each learner writes on a sticky note: one question that today's enzyme research HAS answered for them and one new question that today's research has	Direct learners to the DQB: 'Take one sticky note of each colour. On the green one, write one question that today's research answered for you — be specific, not just yes or no. On the orange one, write one new question that today's work raised in your mind.' [Allow 2 minutes of silent writing —	Two-Column DQB Update (We Now Know / We Still Wonder) — this strategy makes knowledge growth visible and cumulative. By requiring learners to articulate both what has been resolved AND what has been newly opened, the DQB update develops metacognitive awareness of the scientific process	Teacher reads all DQB sticky notes quickly after the lesson (or photographs them). Look for: (1) quality of 'We Now Know' statements — are learners articulating enzyme specificity at a mechanistic level or just restating vocabulary? (2) sophistication of 'We Still Wonder' questions —

Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Human Digestive System (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	RAISED. Learners post both sticky notes on the DQB — answered questions go to a 'We Now Know' column, new questions go to the 'We Still Wonder' column. The teacher reads 3–4 new 'We Still Wonder' questions aloud and briefly acknowledges which will be addressed in upcoming lessons, with particular emphasis on: 'How does the body avoid digesting itself?' (Lesson 5) and 'What happens when enzymes are missing or not working properly?' (Lesson 6 — enzyme deficiency and digestive disease).	circulate and read over shoulders but do not comment yet.] After posting: 'Let me read some of our new wonders aloud.' Read 3–4 of the most scientifically rich 'We Still Wonder' notes. For each one, respond briefly: 'Great question — we will investigate that in Lesson 5' or 'That connects to something real — ulcers in the stomach happen partly because of exactly that question.' Do NOT answer the questions now — preserve the productive tension. Point to the prediction vote tally still visible on the board from the Predict phase: 'Look at our vote from the start of the lesson. How many of you who voted for stomach as the main site would now change your answer? Silent show of hands.' This closes the loop on the opening prediction.	— knowledge generates new questions. The teacher's brief preview of upcoming lessons ('Lesson 5 addresses that') demonstrates that the unit storyline is intentionally structured, motivating learners to stay engaged across lessons. Returning to the opening prediction vote provides a satisfying cognitive closure loop that reinforces the value of evidence-based revision of ideas.	questions about mechanism ('why doesn't salivary amylase work in the stomach?') indicate deeper engagement than recall questions ('what is rennin?'); (3) whether the phenomenon context (ugali, nyama choma) appears in learner language on the notes, indicating genuine connection to the anchoring phenomenon. The DQB data informs the teacher's planning for Lessons 4 and 5.
Model Building Phase 📹 **VIDEO:** Digestive System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Human Digestive System (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve the cumulative alimentary canal diagram model they began in Lesson 1 and annotated in Lesson 2 (drawn in their notebooks or on a shared group sheet). Today, they add a new layer to the model: using their enzyme summary chart as reference, they annotate each region of their diagram with the enzymes active there, the substrate each enzyme acts on, and an arrow showing the direction of chemical change (e.g., Starch → Maltose above the mouth region). Learners also add a colour-coded key: one colour for carbohydrate-digesting enzymes, a second for protein-digesting enzymes, a third for fat-digesting enzymes. This takes 4–5 minutes. Learners then compare their updated model with a partner's and identify one difference to discuss — either a discrepancy to	Say: 'Get out your alimentary canal diagram from Lessons 1 and 2. Today we are adding the enzyme layer — the chemical toolkit layer. Use your enzyme chart to annotate each region. Include: enzyme name, substrate, product, and your colour code for carbohydrate, protein, or fat. You have 4 minutes.' [Set countdown timer. Circulate actively.] After 4 minutes: 'Compare your model with your partner. Find ONE thing that is different between your two models — not wrong necessarily, just different. Discuss for 60 seconds: which version is more accurate and why?' [WAIT 60 seconds.] Cold-call 2 pairs to share their difference: 'Fatuma and Njoroge — what was the difference you found, and how did you resolve it?' [WAIT TIME: 10 seconds.] Close the phase by connecting back to the	Cumulative Annotated Model Revision with Colour Coding — the iterative model-building strategy, used across all lessons in the unit, makes conceptual growth tangible. Each lesson adds a new explanatory layer; learners can literally see their understanding deepen lesson by lesson. The colour-coding system introduces the key concept of enzyme specificity visually — different enzymes belong to different biochemical categories — without requiring learners to memorise this as an abstract rule. Partner comparison activates peer assessment and surfaces residual errors in a low-stakes format. This strategy reflects NGSS Science and Engineering Practice 2 (Developing and Using Models) and supports the KICD core competency of Creativity and Imagination as learners construct	As teacher circulates during model annotation, check for three accuracy indicators: (1) enzymes placed in the correct anatomical region (pepsin must appear at the stomach, not the mouth); (2) substrate-product arrows present and chemically correct (starch → maltose, not starch → glucose at this stage — glucose comes later after further intestinal digestion); (3) colour-coding applied consistently across the diagram. Teacher notes any learner whose model places all enzymes in the stomach — this persistent misconception (identified in the Predict phase vote) needs individual follow-up. The partner-comparison discussion reveals whether learners can evaluate model accuracy using their research evidence, a key science process skill. At close of lesson, teacher asks: 'On a scale of 1–3 —

	resolve or a detail one model has that the other lacks.	phenomenon: 'Your model now shows how the body handles ugali's starch, sukuma wiki's fibre, and nyama choma's protein — each enzyme in the right place, in the right order. In our next lesson, we will test for those nutrients directly. Does your model predict what you expect to find?'	a personalised scientific representation.	hold up fingers — how confident are you that your model now correctly shows where each enzyme acts?' Scan and note the distribution for reflection.

D. TEACHER REFLECTION

1. During the cold-calls in the Explain phase, were learners able to use substrate-product language (e.g., 'starch to maltose') or did they fall back on vague terms like 'breaks it down into smaller pieces'? What does this tell me about the depth of understanding reached, and how should I address vocabulary precision in Lesson 4?

2. Looking at the DQB sticky notes, what proportion of 'We Still Wonder' questions were mechanistic (asking HOW or WHY) versus recall questions (asking WHAT or WHERE)? Does the ratio suggest learners are thinking scientifically, and what instructional adjustments would push more learners toward mechanistic questioning?

3. When I reviewed the enzyme summary charts during the gallery walk, which specific enzyme or region showed the most errors or blank cells across groups? Was this a resource gap (the Kenya Education Cloud material was insufficient), a time gap (12 minutes was too short), or a conceptual gap (learners did not understand the column prompt)? How do I address this before Lesson 4?

4. Did the prediction-vote revisit at the DQB stage produce visible surprise or cognitive shift in learners who had voted for the stomach as the primary digestion site? If learners showed little reaction, what does this suggest about whether the Observe phase evidence was compelling enough — and how might I make the evidence more experiential in future iterations?

5. Was the 40-minute lesson timeline realistic? Which phase ran over, and what was sacrificed? Specifically: did the model-building phase receive enough time (minimum 4 minutes) for learners to meaningfully annotate their diagrams, or was it rushed? How will I adjust pacing in Lesson 4's model-building phase?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)

What did I observe?	Using Kenya Education Cloud resources and printed resource sheets, learners researched the specific enzymes active in each region of the alimentary canal and recorded their findings in region-by-region enzyme summary charts. They observed that different enzymes are produced by different glands/organs (salivary glands, gastric glands, pancreas, intestinal glands), that each enzyme acts on a specific substrate (starch, protein, or fat), and that the number and variety of enzymes increases dramatically from mouth to small intestine.
What did I learn?	The body deploys a precise, region-specific chemical toolkit to digest different food types: salivary amylase begins starch digestion in the mouth; pepsin and rennin break down proteins in the acidic stomach environment; and the small intestine receives pancreatic amylase, lipase, and trypsin from the pancreas, plus peptidases from the intestinal lining itself, completing the digestion of all three macronutrients. Each enzyme is specific to its substrate — salivary amylase cannot digest protein, and pepsin cannot digest starch. This enzyme specificity is a key adaptation of the digestive system.
How does this explain the phenomenon?	This explains the first major puzzle from the anchoring phenomenon: the body does NOT release one general-purpose chemical into the digestive system. Instead, it 'knows' which chemical to deploy at each stage because specific enzymes are produced by specific glands and secreted only at specific locations along the alimentary canal. The ugali's starch, the nyama choma's protein, and the sukuma wiki's fats are each handled by different enzymes in different regions — this regional chemical specialisation is the body's solution to the challenge of digesting a complex, mixed Kenyan meal.

LESSON 4 (40 minutes): Food Tests Practical Part 1 — Detecting Starch and Reducing Sugars in Kenyan Foods	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners carry out the iodine test and Benedict test to identify starch and reducing sugars in familiar Kenyan foods, then link the chemical evidence to the specific enzymes that act on these substrates in the alimentary canal.
Knowledge	Learners will be able to state the reagents, expected colour changes, and food substances detected by the iodine test and Benedict test; identify which Kenyan foods contain starch and which contain reducing sugars; and explain how these findings connect to salivary amylase and pancreatic amylase activity in the body.
Skills	Learners will practise safe laboratory technique, accurate observation and recording of colour changes, tabulation of results, and evidence-based reasoning linking food composition to digestive enzyme specificity.
Attitudes	Learners will demonstrate curiosity, precision, and collaborative responsibility during practical work, and appreciate that everyday Kenyan foods can be investigated scientifically.
Key Inquiry Question	How is the human digestive system adapted to its functions?
Purpose in Storyline	In Lessons 1 to 3 learners predicted, observed, and mapped the chemical toolkit of digestion. Lesson 4 now gives learners hands-on chemical evidence — they test the very foods from the anchoring phenomenon (ugali, sukuma wiki, ripe banana) and confirm which substrates are actually present, generating data that will be used in Lessons 5 through 8 to build a complete picture of digestive adaptation and failure.
Safety Notes	Iodine solution is a mild irritant — learners must avoid contact with eyes and skin and wash hands immediately after use. Benedict reagent and the water bath involve heat — learners must use test-tube holders, keep faces away from the open end of hot test tubes, and never point test tubes at classmates. Broken glassware must be reported to the teacher immediately. No food tested in the lab may be eaten. Lab coats or old shirts and safety goggles (or improvised eye protection) are worn throughout.

B. LESSON OVERVIEW
Learners work in groups of four to perform the iodine test for starch and Benedict test for reducing sugars on five food samples drawn directly from the anchoring phenomenon meal: ugali (cooked and raw maize flour), ripe banana, sukuma wiki extract, and a glucose solution control. They record colour changes in a structured results table, identify which foods contain which carbohydrate type, and explain how their results reveal the substrates on which salivary amylase and pancreatic amylase act. The lesson advances the Driving Question Board by replacing prediction with chemical evidence.

C. LESSON IMPLEMENTATION FRAMEWORK				
Phase	**Learner Experience**	**Teacher Moves**	**Sensemaking Strategy**	**Formative Assessment Strategy**
Predict Phase 📹 **VIDEO:** Dehydration synthesis or a	Each learner receives a printed prediction card showing five food samples in two columns labelled STARCH and REDUCING	Teacher displays the five labelled food samples at the front and says: 'Before we open a single reagent bottle, I want your brain to	Think-Pair-Share followed by a brief whole-class poll (show of hands) to make the distribution of predictions visible. This creates	Teacher scans prediction cards during circulation and notes which learners incorrectly predict that raw maize flour will give a negative

condensation reaction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Roots, Shoots, and W..:Card B: Making Starch Molecules / Brick Starch Directions Source: MIT Blossoms 🔍 Search ARES for similar readings	SUGARS. Working individually for three minutes, learners tick which column they think each food belongs to and write one sentence explaining their reasoning for ugali and one for ripe banana. They then share predictions with their group partner using a Think-Pair-Share so that disagreements surface before testing begins.	go first. Look at these foods from our school lunch puzzle. In your prediction card, decide — does ugali contain starch, reducing sugars, or both? Does a ripe banana? Use what you learned last lesson about salivary amylase and its substrate to guide you.' Teacher circulates, reading over shoulders, and selects two learners with contrasting predictions about the banana to share aloud. Teacher then says: 'Interesting — we have two different predictions about the banana. That is exactly why we do experiments. Let us see whose chemical evidence wins.' Teacher does NOT confirm answers at this stage.	cognitive investment before testing and gives the teacher a live snapshot of prior conceptions about ripening and sugar content.	iodine result — these learners are flagged for targeted questioning during the Observe phase.
Observe Phase 📹 **VIDEO:** Dehydration synthesis or a condensation reaction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Roots, Shoots, and W..:Card B: Making Starch Molecules / Brick Starch Directions Source: MIT Blossoms 🔍 Search ARES for similar readings	Groups collect a pre-prepared tray containing: five labelled test tubes (one per food sample), iodine solution in a dropper bottle, Benedict reagent, a water-bath beaker of hot water (provided by teacher, pre-heated on a kerosene stove or electric hotplate), test-tube holders, a white tile, and a results table worksheet. They carry out both tests in sequence. IODINE TEST: Each group places three drops of each food sample (or a small piece of ugali mashed in water) on the white tile, adds two drops of iodine solution, and records the colour immediately. BENEDICT TEST: Each group places 2 cm3 of each food extract into a clean test tube, adds 2 cm3 of Benedict reagent, places the tube in the hot water bath for five minutes using a test-tube holder, and records the final colour. All results are entered into the structured table with columns: Food Sample, Iodine Test Colour,	Teacher gives a live demonstration at the front before groups begin, narrating each step aloud: 'I am adding exactly two drops of iodine to the ugali sample on my white tile — watch the colour change now — that blue-black colour is your positive result for starch. If the tile stays brown-orange, starch is absent.' Teacher then demonstrates Benedict test safety: 'Notice I am holding the test tube at an angle, pointing away from me, using the holder. The tube is hot — I am not using my bare hand.' During group work, teacher circulates using the following targeted questions — to any group whose banana Benedict result is orange-red: 'What colour tells you reducing sugar is present? What sugar do you think is in a ripe banana — and why might an unripe banana give a different result?' To any group confused about why raw maize flour also turns blue-black: 'Salivary amylase	Structured observation table forces learners to convert continuous colour observations into categorical evidence (Yes or No), mirroring how scientists report qualitative data. The side-by-side comparison of cooked ugali versus raw maize flour is designed to surface the concept that cooking gelatinises starch but does not remove it.	Teacher checks each group results table before they move to the Explain phase. Groups with incorrect categorical entries (for example, marking glucose solution as starch-positive) are asked to repeat that single test with teacher supervision before proceeding.

	Starch Present (Yes or No), Benedict Test Colour, Reducing Sugar Present (Yes or No).	breaks starch into maltose — does that mean starch has disappeared from the maize before you chew it? What does your iodine result tell you about raw maize?' Teacher uses a minimum 15-second wait time after each question before accepting a response. Timer is visible on the board.		
Explain Phase 📹 **VIDEO:** Dehydration synthesis or a condensation reaction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Roots, Shoots, and W..:Card B: Making Starch Molecules / Brick Starch Directions Source: MIT Blossoms 🔍 Search ARES for similar readings	Using their completed results table, each group writes a group explanation paragraph answering three prompt questions printed on the worksheet: (1) Which foods from our school lunch contain starch, and which enzyme in the body would begin digesting that starch first? (2) Why does ripe banana give a positive Benedict result — what has happened to the starch inside the banana? (3) Why is it important that salivary amylase acts on starch in the mouth before the food even reaches the stomach? Groups then nominate one spokesperson to share their answer to one prompt with the class.	Teacher cold-calls three group spokespeople — one per prompt question — without advance warning. Before calling on the first spokesperson, teacher says: 'I am going to give you fifteen seconds of silence right now to make sure your group agrees on your answer to prompt one.' Teacher counts silently, then cold-calls. After the first group shares, teacher uses probing language: 'Can another group add to that, or respectfully challenge it?' For prompt three (the mouth-to-stomach link), teacher explicitly connects back to the phenomenon: 'Think about our ugali puzzle. When a learner swallows ugali, the starch digestion has already begun in the mouth — so by the time ugali reaches the small intestine, some starch has already become maltose. How does that change what enzymes the small intestine needs to release?' Teacher uses a further 15-second wait before accepting responses. Teacher closes the Explain phase by writing on the board: EVIDENCE FROM FOOD TEST — UGALI CONTAINS STARCH — SUBSTRATE FOR SALIVARY AMYLASE AND PANCREATIC AMYLASE, and asking learners to copy this into their notebooks as a key evidence statement.	The three-prompt paragraph format scaffolds learners from raw data to mechanism to function — moving up Bloom's taxonomy from recall (which test gave which colour) to analysis (why ripe banana differs from unripe) to application (why mouth digestion matters for the whole system). Cold-calling with wait time ensures that explanation is distributed across the class rather than dominated by confident learners.	Teacher listens for two key misconceptions during cold-call responses: (a) that starch and sugar are the same substance, and (b) that a negative Benedict result for ugali means ugali has no sugar at all. Both misconceptions are addressed in real time by redirecting groups back to their enzyme summary chart from Lesson 3.

Driving Question Board (DQB) Creation 📹 **VIDEO:** [Dehydration synthesis or a condensation reaction] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Roots, Shoots, and W..:Card B: Making Starch Molecules / Brick Starch Directions Source: MIT Blossoms 🔍 Search ARES for similar readings	In the final five minutes of the lesson, each group writes one new sticky note (or a strip of paper) to add to the class DQB. The note must begin with one of three sentence starters: WE NOW KNOW... / WE CAN NOW EXPLAIN... / WE STILL WONDER... Groups are encouraged to use their food test evidence directly. Example contributions anticipated: WE NOW KNOW ugali contains starch and that salivary amylase is the first enzyme to act on it. WE STILL WONDER what happens to the starch that reaches the small intestine without being fully broken down — does it get wasted? WE CAN NOW EXPLAIN why ripe bananas taste sweeter than unripe ones — the starch has been converted to reducing sugars.	Teacher reads two or three DQB contributions aloud, highlighting how today's chemical evidence has moved the class from prediction to proof. Teacher then bridges to the next lesson: 'Next lesson we test for proteins and fats — the nyama choma and the cooking fat from our sukuma wiki. By the end of Lesson 5, we will have chemical evidence for all three macronutrients in our school lunch, and we can begin explaining how the whole digestive system is adapted to handle each one.' Teacher leaves the DQB visible on the wall so learners can add questions between lessons.	The three sentence starters structure metacognitive reflection — learners must distinguish between confirmed knowledge, explanatory progress, and remaining uncertainty. This models scientific thinking and keeps the storyline momentum visible to all learners.	Teacher scans DQB additions to check that groups are connecting food test results to enzyme specificity rather than leaving them as isolated observations. Any note that mentions only colour change without linking to an enzyme or substrate is returned to the group with the prompt: 'Which body chemical acts on what you just tested? Add that link to your note.'
Model Building Phase 📹 **VIDEO:** Dehydration synthesis or a condensation reaction Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **PDF:** Roots, Shoots, and W..:Card B: Making Starch Molecules / Brick Starch Directions Source: MIT Blossoms 🔍 Search ARES for similar readings	This phase is embedded within the Observe phase practical rather than as a separate model-building activity in Lesson 4 — learners construct a visual data model (the results table) and a written explanatory model (the group paragraph) rather than a physical 3D model, which is reserved for Lesson 7. As a brief closing consolidation, each learner individually draws a simple two-box diagram in their notebook: Box 1 labelled MOUTH with one sentence describing what happens to the starch in ugali there; Box 2 labelled SMALL INTESTINE with one sentence describing what happens to any remaining starch. An arrow between boxes is labelled with the product leaving the mouth (maltose and remaining starch).	Teacher models the two-box diagram on the board using the ugali example, deliberately making one error — writing LIPASE instead of PANCREATIC AMYLASE in the small intestine box — and asking the class: 'Check my diagram against your enzyme chart from Lesson 3. Has anyone spotted something that needs correcting?' Teacher waits 15 seconds. After a learner identifies the error, teacher corrects it visibly and says: 'That error is exactly the kind of mistake that a doctor could not afford to make when explaining why a patient is not digesting carbohydrates properly. Precision in enzyme names matters.' This connects to career awareness embedded in the unit storyline.	The deliberate teacher error activates critical evaluation rather than passive copying. The two-box diagram begins the visual modelling work that will scale up into the full 3D digestive system model in Lesson 7, providing continuity across the storyline.	Learners hold up their two-box diagrams for a quick gallery check before leaving the lab. Teacher looks for correct enzyme names and correct product labelling on the arrow. Any diagram missing the arrow label is flagged with a written question mark and returned for correction at the start of Lesson 5.

D. TEACHER REFLECTION
After the lesson, consider the following questions: Did all groups complete both food tests within the available practical time, and if not, which procedural step caused the delay — could pre-prepared extracts reduce setup time next time? Which misconception was most common during cold-calling — starch-equals-sugar confusion or incorrect enzyme-region matching — and how will you address it explicitly at the start of Lesson 5? Did the 15-second wait time feel comfortable, or did you intervene too early — note one moment where extended wait time produced a stronger learner response than you expected. Were safety procedures followed consistently by all groups, and is there any learner who needs additional coaching on hot water bath technique before the Lesson 5 Biuret practical involving NaOH?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed that ugali (both cooked and raw maize flour) turned blue-black with iodine solution, confirming the presence of starch; that ripe banana and glucose solution turned brick-red with Benedict reagent, confirming reducing sugars; and that sukuma wiki extract gave weak or negative results for both tests, which raised a new question about what nutrients sukuma wiki actually provides.
What did I learn?	Learners learned that the iodine test and Benedict test are reliable, reproducible chemical tests for specific carbohydrate types; that starch and reducing sugars are chemically different and require different enzymes (amylase for starch, other enzymes for resulting sugars); and that the ripening of a banana involves conversion of starch to reducing sugars, linking biology to the chemistry of everyday Kenyan foods.
How does this explain the phenomenon?	Learners explained that the starch in ugali is the primary substrate for salivary amylase in the mouth and pancreatic amylase in the small intestine; that the reducing sugars produced by amylase action represent the product of successful carbohydrate digestion; and that their food test evidence directly supports the enzyme-specificity principle established in Lesson 3 — each enzyme acts on a specific substrate, and the substrates are chemically real and detectable in the actual foods of the anchoring phenomenon meal.

LESSON 5 (40 minutes): Food Tests Practical (Part 2) — Detecting Proteins and Fats in the Kenyan Meal	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners use the Biuret test and ethanol emulsion test to confirm the presence of proteins and fats in components of the Ugali and Sukuma Wiki meal, linking chemical evidence to the stomach's pepsin action and the small intestine's fat emulsification.
Knowledge	– Describe the Biuret test procedure and interpret a violet/purple colour change as a positive result for proteins – Describe the ethanol emulsion test procedure and interpret a milky-white emulsion as a positive result for fats – Explain how the presence of protein in nyama choma links to the role of pepsin (a protease) secreted in the stomach – State why fat digestion (lipase + bile salts) is slower and occurs primarily in the small intestine
Skills	– Safely handle NaOH and ethanol, following correct laboratory safety protocols – Accurately perform the Biuret and ethanol emulsion food tests, record colour changes, and compare results across groups – Construct a data table and draw evidence-based conclusions about nutrient composition of meal components
Attitudes	– Appreciate that chemical tests provide objective, reproducible evidence — not guesswork — about what nutrients are present in familiar Kenyan foods – Value safety and precision as non-negotiable in scientific investigation
Key Inquiry Question	How is the human digestive system adapted to its functions?
Purpose in Storyline	Having confirmed starch and reducing sugars in Lessons 4, learners now complete the nutrient-detection evidence base by identifying proteins and fats in nyama choma, bean broth, milk, butter, cooking oil, and sukuma wiki extract — giving them the full nutrient profile of the anchoring meal and raising the question of why different enzymes are needed for each nutrient class.
Safety Notes	NaOH (sodium hydroxide) used in Biuret reagent is corrosive — learners must wear safety goggles and nitrile gloves; any spill on skin should be flushed immediately with large amounts of water for at least 10 minutes. Ethanol is flammable — no open flames in the laboratory during the ethanol emulsion test; use only cold water baths if heating is needed. Teacher must demonstrate safe pipetting technique before learners handle reagents. Dispose of NaOH waste into the designated neutralisation bucket, not the sink. Teacher must have a first-aid kit and eye-wash station accessible.

B. LESSON OVERVIEW

This lesson is the second of two food-test practicals in the evidence-gathering sequence. Learners enter having already confirmed the presence of starch (iodine test) and reducing sugars (Benedict's test) in ugali and sukuma wiki in the preceding lesson. The remaining puzzle is: what about the nyama choma (protein) and the cooking fat used in the meal? Today learners perform two classical biochemical tests — the Biuret test for proteins and the ethanol emulsion test for fats — on a carefully chosen set of food samples that mirror the anchoring phenomenon: nyama choma extract, bean broth (from githeri), milk (a familiar protein-and-fat source), butter, cooking oil, and a sukuma wiki leaf extract. By the end of the practical, every group has a complete evidence table covering all four major nutrient classes across the Kenyan meal.

The lesson is explicitly designed to bridge from chemical detection back to physiology. Once learners confirm that nyama choma gives a strong violet Biuret result, the teacher uses this as the entry point to revisit pepsin: the stomach secretes pepsin precisely because the meal contains protein, and the low pH of gastric acid is needed to activate pepsinogen into pepsin. Similarly, confirming fat in cooking oil and butter primes learners to ask why fat digestion requires both bile salts (from the liver, stored in the gall bladder) and lipase — and why it happens much later, in the small intestine, than carbohydrate digestion. These connections advance the driving question by showing that the body's chemical toolkit is matched, with remarkable specificity, to the nutrients present in the meal.

Safety is foregrounded throughout this lesson because both key reagents present real hazards. The teacher models correct NaOH handling and pipetting before learners touch the Biuret reagent, and the ethanol emulsion test is performed with strict no-open-flame protocols. The PCIs of Safety and Security (KICD) are thus embedded as genuine laboratory practice rather than as an afterthought, and the value of Unity is reinforced as group members divide safety roles (designated goggles-checker, spill-watcher, recorder).

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** 2015 AP Chemistry free response 2a (part 1 of 2) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Test Your Strength (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown a photograph (or the actual samples, covered) of the five test substances: nyama choma extract, bean broth, a small cup of milk, a small piece of butter, cooking oil, and a sukuma wiki leaf extract. Without touching or testing anything, each learner individually writes on a sticky note or in their exercise book: (a) which samples they predict will test positive for protein, (b) which will test positive for fat, and (c) one reason for each prediction. Learners then share predictions in pairs (Think-Pair-Share) before two pairs report to the class.	Display the six labelled sample containers prominently. Say: 'Before we do a single test, I want you to think like a detective. Look at these samples from our Ugali and Sukuma Wiki meal. Write down — individually, no discussing yet — which ones you think contain protein and which contain fat, and why you think that.' Allow 2 minutes of silent individual writing. Then say: 'Now share your prediction with your partner. You have 90 seconds.' After pair discussion, cold-call 4 learners (aim for gender balance across the class): 'Amina — what did you and your partner predict about nyama choma, and what was your reasoning?' WAIT TIME: minimum 15 seconds after asking each learner before accepting the answer or redirecting. Record the class predictions on the board in a simple two-column table (Protein positive / Fat positive) so predictions can be revisited after the practical. Do not confirm or deny any prediction yet — say: 'Excellent reasoning. Let us see whether the chemistry agrees with you.'	Predict-Before-Observe (Anchored Prediction): Learners activate prior knowledge about macronutrient sources (built in Lesson 3) and commit to a public prediction. Recording predictions on the board creates accountability and cognitive dissonance when results differ from expectations — a key driver of genuine sensemaking.	Teacher scans the room during individual writing: do learners correctly identify nyama choma and milk as protein sources, and butter/cooking oil as fat sources? Misconception to watch for: learners predicting sukuma wiki is negative for fat (it actually contains trace lipids, producing a faint emulsion) or that milk contains no fat (it contains both). Note which learners are uncertain — these are the cold-call targets in the Explain Phase.
Observe Phase	Learners work in groups of 4. Each group has a set of six labelled test	Before learners touch any reagent, say: 'I am going to demonstrate	Structured Group Investigation with Whole-Class Data Collation:	Teacher circulates and checks: (1) Are learners adding reagents in

📹 **VIDEO:** [2015 AP Chemistry free response 2a (part 1 of 2)] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Test Your Strength (PLIX)] Source: CK-12 🔍 Search ARES for similar readings	tubes for the Biuret test and a second set of six for the ethanol emulsion test. For the Biuret test: learners add 2 cm³ of each test sample, then add 2 cm³ of dilute NaOH solution, mix gently, then add 4–5 drops of 1% copper sulphate solution ($CuSO_4$) and observe. For the ethanol emulsion test: learners add 2 cm³ of each sample to a test tube, add 5 cm³ of absolute ethanol and shake to dissolve any fat, then pour the solution into a test tube containing 5 cm³ of cold water and observe for a milky-white emulsion. All colour changes and observations are recorded immediately in a group results table. Groups then share results with the class by calling out readings while the teacher completes a whole-class collated results table on the board.	the safety steps for NaOH first — everyone watches.' Demonstrate: putting on goggles and gloves, using a dropper to transfer NaOH, keeping the dropper tip away from the face. Then say: 'I also need everyone to check: is there any flame or spirit lamp burning on your bench? Before you add ethanol to anything, all flames on your bench must be out. Your bench safety monitor — the person sitting closest to the window — is responsible for checking this.' Circulate during the practical; pause at each bench to ask: 'What colour are you getting? What does that tell you?' After all groups finish and the whole-class table is on the board, say: 'Look at the table. Is there anything surprising compared to the predictions we made at the start?' WAIT TIME: 15 seconds. Cold-call 2 learners who had different predictions from the results. For the collation stage, explicitly model data recording: 'We write what we actually see — violet, no colour change, milky emulsion — not what we hoped to see. That is scientific integrity.'	Each group generates independent data, then the class pool reveals consistency (or anomalies) across groups. Collating results on a single board table makes the evidence communal and publicly interrogable — a core science-and-engineering practice (Analysing and Interpreting Data, NGSS SEP 4). Anomalous group results become productive discussion points rather than errors to hide.	the correct order for Biuret (NaOH first, then $CuSO_4$)? Incorrect order gives a false blue rather than violet. (2) Are learners recording actual observations (colour, turbidity) rather than just writing 'positive/negative'? (3) Does the class results table show agreement across groups for the strong positives (nyama choma, milk for Biuret; butter, cooking oil for ethanol emulsion)? Disagreement on sukuma wiki fat result is expected and productive — use it in the Explain Phase.
Explain Phase 📹 **VIDEO:** [2015 AP Chemistry free response 2a (part 1 of 2)] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** [Test Your Strength (PLIX)] Source: CK-12	Learners return to their seats with their completed results tables. The teacher leads a whole-class discussion structured around three questions written on the board, which learners answer first in writing (2 minutes), then discuss. The three questions are: (1) Why does nyama choma turn violet with Biuret reagent — what does violet mean at the molecular level? (2) How does this link to what pepsin does in the stomach? (3) Why does fat digestion not begin in the mouth or stomach the way starch	Write the three questions on the board before starting discussion. Say: 'Take two minutes — in silence — to write your answers to all three questions. Use what you already know from Lessons 2 and 3 to help you.' Allow 2 full minutes of silent writing. Then say: 'Let us tackle question one. The Biuret reagent turns violet in the presence of protein. What is it actually reacting with?' WAIT TIME: 15 seconds. Cold-call Learner 1 (a mid-confidence learner). If the answer is	Evidence-to-Mechanism Reasoning (NGSS SEP 6 — Constructing Explanations): Learners are not just interpreting colour changes; they are using the colour changes as evidence to explain why specific physiological mechanisms exist. The three-question scaffold moves learners from observation (violet colour) → mechanism (peptide bond coordination) → physiological consequence (pepsin, low pH, lipase, bile). The sentence-completion exit move consolidates	Teacher listens for: (1) Can learners connect the violet Biuret result to peptide bonds, not just 'protein is present'? (2) Do learners correctly name pepsin and the role of HCl/low pH in activating it? (3) Do learners articulate why fat digestion is delayed (emulsification prerequisite, insolubility of fats)? Misconception to correct: 'bile digests fat' — bile emulsifies fat; lipase digests it. The sentence-completion responses give a written formative check that can be

🔍 Search ARES for similar readings	digestion does? Learners annotate their results tables with explanatory notes as the discussion proceeds. At the end of the Explain Phase, learners write one sentence completing the stem: 'The body needs different enzymes for proteins and fats because…'	incomplete, say: 'Good start — who can add to that?' Cold-call Learner 2. Guide toward: the copper ions in $CuSO_4$ coordinate with the peptide bonds in protein chains, producing the characteristic violet colour — more peptide bonds, stronger violet. For question 2, say: 'We confirmed that nyama choma is full of protein. Your stomach is about to receive that nyama choma. What enzyme does it use, and what conditions does it need?' WAIT TIME: 15 seconds. Cold-call Learner 3. Prompt if needed: 'Think back to Lesson 3 — which enzyme is secreted in the stomach, and why does it need such a low pH?' For question 3, say: 'Fat gave us a positive emulsion test. But we know bile salts and lipase are needed for fat digestion — and both are released into the small intestine, not the stomach. Why do you think fat digestion is delayed?' WAIT TIME: 15 seconds. Cold-call Learner 4. Draw out: fat is insoluble in water, so it must be emulsified by bile salts before lipase can access it; this structural challenge means fat digestion requires more steps and more time than protein digestion. Connect explicitly to the driving question: 'So our Biuret result with nyama choma is direct chemical evidence that the stomach's pepsin-acid system exists precisely because this meal contains protein. The chemistry of the food drives the chemistry of the gut.'	the key conceptual link.	reviewed after class.
Driving Question Board (DQB) Creation 📹 **VIDEO:**	Each group receives two sticky notes of different colours (e.g., green = new understanding, orange = new question). On the green note, groups write one	Direct learners to the DQB: 'Our board has been growing since Lesson 1. Today we have chemical evidence for proteins and fats. Take two sticky notes —	Public Knowledge Tracking (DQB Update): Making new understanding and new questions visible on a shared board externalises the class's growing	Teacher reads all posted green notes after class: do they reflect accurate conceptual gains (e.g., protein → pepsin → low pH; fat → bile emulsification → lipase) or

[2015 AP Chemistry free response 2a (part 1 of 2)] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Test Your Strength (PLIX) Source: CK-12 🔍 Search ARES for similar readings	statement of something they can now explain about the driving question that they could not explain before this lesson. On the orange note, groups write one new question that today's results raised — for example: 'If the stomach digests protein, why doesn't it digest itself?' or 'How exactly does bile break up fat — is it like soap?' Groups stick their notes onto the class Driving Question Board under the headers 'We now understand…' and 'We still wonder…'. The teacher reads two or three notes aloud and connects them to upcoming lessons.	green and orange. On the green: what can you now explain about how the body handles the nyama choma or the cooking fat in our meal? On the orange: what new question did today's practical raise for you?' Allow 3 minutes for group writing. As groups post their notes, scan quickly. Read aloud one green note: e.g., 'This group writes: we can now explain why the stomach needs an acidic environment — it has to activate pepsin to break down the protein in nyama choma. That is exactly right.' Then read one orange note: e.g., 'This group asks: if pepsin digests protein, why doesn't the stomach wall get digested? That is a brilliant question — it is exactly what the mucus lining of the stomach is for, and we will return to this in Lesson 7 when we discuss what happens when that system fails.' Connect the self-digestion question explicitly to the driving question: 'Remember — one of our original puzzles was how the body avoids digesting itself. Today's practical has given you the chemical evidence that makes that question urgent.'	model of digestion. The two-colour system forces learners to distinguish between consolidated knowledge and productive uncertainty — both are valued, which creates a psychologically safe environment for genuine inquiry. Connecting orange questions to future lessons shows learners that their questions have a place in the storyline.	merely procedural gains ('we learned how to do the Biuret test')? Procedural-only green notes indicate the Explain Phase did not achieve the desired mechanistic depth and should inform the opening of Lesson 6. Orange questions that reveal the self-digestion puzzle are a strong indicator of genuine engagement with the driving question.
Model Building Phase 📹 **VIDEO:** 2015 AP Chemistry free response 2a (part 1 of 2) Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Test Your	Learners retrieve the cumulative diagram of the alimentary canal they have been building since Lesson 2. Using a coloured pen (a different colour from those used in previous lessons), they add two new layers of annotation: (1) at the stomach, they draw an arrow and label it 'pepsin (protease) activated by HCl — digests protein from nyama choma/bean broth → peptides'; (2) at the small intestine, they draw an arrow and label it 'bile salts (emulsify fat) + lipase → fatty acids + glycerol — from	Say: 'Take out your cumulative digestive system diagram. Every lesson, we have been adding to this model as we gather more evidence. Today, your Biuret result is your evidence that protein is present in the meal — and your ethanol emulsion result is your evidence that fat is present. Now add those facts to your model.' Circulate and check that learners are annotating the correct anatomical location: pepsin at the stomach, lipase/bile at the small intestine. If a learner writes lipase	Cumulative Model Revision (NGSS SEP 2 — Developing and Using Models): The running diagram is the class's shared explanatory model of the phenomenon. Each lesson's new evidence adds a layer, making the model progressively more complete and more powerful. The written sentence forces learners to articulate the connection between the chemical test result (observation) and the physiological mechanism (explanation) — closing the loop between the	Teacher collects or photographs three to four learner diagrams at the end of the lesson. Check: (1) Is pepsin annotated at the stomach (not the mouth or small intestine)? (2) Are bile salts and lipase both annotated at the small intestine? (3) Does the written sentence connect the Biuret/emulsion results to physiological function rather than simply restating the test result? Common error: confusing bile with an enzyme — correct this now before it solidifies ahead of Lesson 7.

Strength (PLIX) Source: CK-12 🔍 Search ARES for similar readings	cooking fat/butter'. Below their diagram, learners write one sentence answering: 'How does today's chemical evidence (Biuret = violet; ethanol emulsion = milky) change or confirm your model of what happens to the nyama choma and cooking fat after they are swallowed?'	at the stomach, say: 'Look at your Lesson 3 notes — where exactly is lipase secreted, and what has to happen to the fat first before lipase can act on it?' After 4 minutes, ask one learner to share their annotation aloud. Say: 'Notice how your model is getting richer each lesson. In Lesson 1 you could not explain how the body knew what chemical to release at each stage. Can you answer that question better now?' WAIT TIME: 15 seconds. Take one volunteer response. Close the lesson by saying: 'By next lesson, we will have a complete nutrient map of the Ugali and Sukuma Wiki meal — and we will use it to ask what happens when any one of these systems fails.'	Observe and Explain phases.	

D. TEACHER REFLECTION
1. During the Biuret test, did all groups obtain a clear violet result for nyama choma and milk, and a clear negative for ugali? If any group obtained an unexpected result, what procedural error (wrong order of reagent addition, contaminated equipment, insufficient sample concentration) most likely caused it — and how will I address this before the next food-test lesson? 2. When I cold-called four learners during the Explain Phase, did learners correctly distinguish between bile (emulsification) and lipase (chemical digestion of fat)? If not, at what point in Lesson 6 should I revisit this distinction before moving on to absorption? 3. Did the orange sticky notes on the DQB reveal that learners have genuinely grasped the self-digestion puzzle ('how does the stomach avoid digesting itself?'), or did the questions remain at a surface level ('why does Biuret turn violet?')? What does this tell me about the depth of conceptual engagement during the Explain Phase? 4. Safety compliance: did all groups correctly wear goggles and gloves for the NaOH handling, and did all flames go out before ethanol was opened? If any safety lapse occurred, how should I restructure the safety briefing in future practicals — and should I involve the class Safety Monitor role more explicitly? 5. Looking at the cumulative diagrams collected at the end of the lesson, are learners building a spatially accurate model of the alimentary canal (correct sequence: mouth → oesophagus → stomach → small intestine → large intestine) or is the anatomy still confused? If spatial confusion persists, should Lesson 6 begin with a brief anatomy orientation before moving to absorption?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed that nyama choma extract and milk turned violet with Biuret reagent (positive for protein), while ugali extract remained blue (negative). Butter and cooking oil produced a milky-white emulsion with the ethanol emulsion test (positive for fat), while sukuma wiki gave a faint emulsion and ugali gave none. These results were consistent across all class groups.
What did I learn?	Proteins contain peptide bonds that coordinate with copper ions in Biuret reagent to produce a violet colour. Fats dissolve in

	ethanol and form a milky emulsion when the ethanol solution is poured into water. The presence of protein in nyama choma explains why the stomach secretes pepsin (a protease) activated by hydrochloric acid. The presence of fat explains why bile salts (emulsification) and lipase (chemical digestion) are both needed in the small intestine — fat digestion is delayed compared to starch and protein digestion because fat must be emulsified first.
How does this explain the phenomenon?	The Ugali and Sukuma Wiki meal contains all four major nutrient classes: starch (confirmed in Lesson 4), reducing sugars (Lesson 4), protein (today's Biuret test on nyama choma and bean broth), and fat (today's ethanol emulsion test on cooking oil and butter). This explains part of the puzzle: the body 'knows' which chemical to release at each stage because each region of the alimentary canal is matched to the specific nutrient chemistry present — salivary amylase in the mouth for starch, pepsin in the stomach for protein, and bile plus lipase in the small intestine for fat. The meal's nutrient profile drives the digestive system's regional specialisation.

LESSON 6 (40 minutes): Adaptations of the Human Digestive System — Structure Meets Function	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners explain how the structural adaptations of the small intestine, stomach, and sphincters are precisely matched to their digestive functions, resolving the puzzle of how the narrow small intestine absorbs all nutrients from a Kenyan meal.
Knowledge	– Describe the structural adaptations of the small intestine (villi, microvilli/brush border, long length, rich blood supply) and explain how each adaptation increases efficiency of absorption. – Explain how rugae in the stomach wall allow the stomach to expand and increase surface area for mechanical and chemical digestion of nyama choma protein. – Describe the role of the cardiac sphincter (preventing acid reflux) and pyloric sphincter (controlling chyme release) in regulating digestion. – Explain how the stomach avoids self-digestion through the mucus lining and the regulation of pepsin as an inactive zymogen (pepsinogen).
Skills	– Annotate a labelled diagram of the digestive system, linking each structural feature to its specific function using evidence gathered in Lessons 1–5. – Construct and present a structured argument connecting structural adaptation to function during a gallery walk.
Attitudes	– Appreciate that the human body's digestive machinery is a finely tuned system where structure and function are inseparable, inspiring curiosity about biological design. – Respect peers' annotated diagrams and contributions during the gallery walk, recognising multiple valid ways of representing the same knowledge.
Key Inquiry Question	How is the human digestive system adapted to its functions?
Purpose in Storyline	This EXPLAIN lesson is the conceptual payoff of the evidence-gathering sequence: learners now have enzyme knowledge (Lesson 3), food composition data (Lessons 4–5), and mechanical digestion understanding (Lesson 2) to finally answer why the small intestine — a tube barely 3 cm wide — can absorb all the nutrients from ugali, sukuma wiki, and nyama choma, and why the stomach does not digest itself.
Safety Notes	No chemical or laboratory hazards. Learners handle printed diagrams and photomicrographs only. Ensure printed images are age-appropriate. If digital devices are used to view photomicrographs, supervise internet use per school ICT policy.

B. LESSON OVERVIEW

Lesson 6 is the pivotal EXPLAIN lesson in the unit storyline. After five lessons of evidence-gathering — observing the anchoring phenomenon, investigating mechanical and chemical digestion, mapping enzyme action by region, and testing for nutrients using food tests — learners now turn to the question of structure. The driving question asks how a single alimentary canal can process the chemically diverse components of a Kenyan school lunch (starch from ugali, fibre and micronutrients from sukuma wiki, and protein and fat from nyama choma) without error, without self-destruction, and within hours. The answer lies not just in chemistry but in anatomy: the digestive system's structures are exquisitely adapted to their functions.

In this lesson, learners examine labelled diagrams and photomicrographs of three key structural features: (1) the villus and brush-border structure of the small intestinal lining, which multiplies absorptive surface area by up to 600-fold; (2) the rugae of the stomach wall, which allow the stomach to stretch and maximise contact with food during churning; and (3) the cardiac and pyloric sphincters, which act as precision valves controlling the movement of food between regions. Learners construct annotated diagrams, connecting each feature to its function and to specific moments in the digestion of the Kenyan meal. A gallery walk allows groups to share one adaptation each, building a whole-class explanatory picture.

The lesson closes by returning to the Driving Question Board: learners identify which of their original questions — particularly "how does the body know which chemical to release?" and "how does the stomach avoid digesting itself?" — can now be answered with structural and chemical evidence. The model is updated to incorporate structural annotations, preparing learners for Lesson 7 (disorders of the digestive system) where these same adaptations will be examined in failure mode.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Centripetal force problem solving Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Surface Area and Nets (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners are shown a single striking image: a cross-section photomicrograph of the small intestinal wall showing finger-like villi alongside a photograph of the inside of a stomach showing rugae folds. Without any labels, learners write a two-sentence prediction in their notebooks: 'What do you think these structures DO, and why do you think the small intestine needs them to absorb nutrients from ugali?' They then share predictions with their shoulder partner before two pairs are cold-called to share with the class.	Display the two images side by side on the board (or distribute printed A4 sheets if no projector is available). Say: 'Before I tell you anything about these structures, I want you to think like a design engineer. Look at these two images from inside a human body — one from the small intestine, one from the stomach. Write in your notebook: what do you think these shapes are for, and why would the small intestine need them when digesting our ugali and nyama choma?' Allow 2 minutes of silent individual writing. Then say: 'Share your prediction with your partner.' Allow 90 seconds. After partner talk, say: 'Let us hear two predictions.' Cold-call 2 pairs (not volunteers). Record key prediction words on the board under the heading 'Our Predictions.' Do NOT confirm or correct — say: 'We will test these predictions against evidence in the next phase.' WAIT TIME: minimum 12 seconds after posing the initial question before permitting any sharing.	Elicit-Before-Explain (Activate Prior Knowledge): By requiring learners to commit a prediction to writing before instruction, the teacher surfaces existing mental models about structure and function. Recording predictions publicly on the board creates accountability and gives learners a stake in the subsequent evidence phase. This strategy is grounded in NGSS Science and Engineering Practice 6 (Constructing Explanations) — learners begin building an explanation before they have all the evidence, which they will then refine.	Teacher scans written predictions while circulating during the 2-minute writing window. Look for: (1) Do learners connect shape to function (e.g., 'folds increase surface area') or do they describe shape only (e.g., 'it looks bumpy')? (2) Do any learners already reference absorption or nutrient uptake? Learners who connect shape to function demonstrate prior schema for structure-function relationships and can be used as anchor voices during the Explain Phase. Learners who describe shape only need scaffolding toward functional language in the Observe Phase.
Observe Phase 📹 **VIDEO:**	In groups of four, learners receive a resource pack containing: (A) a clearly labelled diagram of a villus	Distribute resource packs. Say: 'You have eight minutes. Your only job right now is to be a careful	Structured Observation with Evidence Tables (NGSS Practice 3 — Planning and Carrying Out	Teacher checks Evidence Tables during circulation. Target indicators: (1) Do learners record

Centripetal force problem solving Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Surface Area and Nets (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	cross-section showing epithelial cells, microvilli (brush border), lacteal, capillary network, and goblet cells; (B) a photomicrograph or high-quality digital image of rugae in the stomach wall; (C) a diagram of the stomach showing the positions of the cardiac sphincter (oesophageal junction) and pyloric sphincter (duodenal junction); and (D) a two-column Evidence Table (Structure | What I Observe About Its Shape/Features). Learners spend 8 minutes completing the Evidence Table — observing and recording structural features without yet explaining them. Groups are instructed: 'Describe only what you see — shapes, textures, positions, quantities. Do not write function yet.'	observer — like a scientist examining tissue samples in a lab at Kenyatta University. Fill in the left column of your Evidence Table: describe what you can see about the shape and features of each structure. Do NOT write function yet — that comes next.' Circulate continuously, pausing at each group for 60–90 seconds. Use probing questions: 'How many villi do you estimate per square millimetre in image A? What does that suggest about the total number in a 6-metre intestine?' and 'What happens to the stomach's internal volume if the rugae unfold? Can you estimate from the diagram?' At the 6-minute mark, give a 2-minute warning: 'Two minutes — make sure you have at least three observations for each of the three structures.' Do NOT provide answers. If a group fills in the function column early, say: 'Good thinking — park that in your mind and write it in pencil in the margin. We will use it in the Explain Phase.' WAIT TIME: When asking probing questions, wait 12 seconds before offering a hint.	Investigations / Practice 8 — Obtaining and Evaluating Information): Separating observation from explanation is a disciplinary practice central to science. By constraining learners to description only in this phase, the teacher builds the habit of distinguishing data from interpretation. The Evidence Table becomes a shared artefact that groups will draw on directly in the Explain Phase, making the transition between phases explicit and traceable.	quantitative or semi-quantitative observations (e.g., 'many finger-like projections', 'folds cover approximately half the stomach wall') rather than vague descriptors only? (2) Do learners notice the capillary network inside the villus diagram — this is a key structural detail needed for the Explain Phase. (3) Are groups identifying both sphincters correctly by position? Groups missing the capillary network in the villus should be prompted: 'Look inside the villus itself — what else do you see besides the outer cells?'
Explain Phase 📹 **VIDEO:** Centripetal force problem solving Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Surface Area and Nets (Flexbook) Source: CK-12 🔍 Search ARES for similar	Each group receives an Annotated Diagram Template: an unlabelled outline of the small intestinal wall (showing villi at two scales: tissue level and cellular/brush-border level), the stomach interior, and a simplified alimentary canal diagram showing sphincter positions. Using their Evidence Tables, learners annotate each structure — drawing arrows and writing function statements in their own words. They must connect each annotation explicitly to the Kenyan meal: e.g., 'Microvilli on	Before groups begin annotating, conduct a 3-minute whole-class structured discussion using the following sequence. First, project or draw a simple villus diagram. Ask: 'From your observations, what single structural feature of the small intestine most increases its surface area?' WAIT 12 seconds. Cold-call 3 learners. After responses, provide the bridging explanation: 'Scientists estimate that without villi and microvilli, the small intestine would need to be approximately 2,000	Structured Academic Controversy → Annotated Diagram → Gallery Walk (NGSS Practice 6 — Constructing Explanations; Practice 7 — Engaging in Argument from Evidence): The lesson moves from whole-class discussion (building shared vocabulary and key concepts) to individual group annotation (consolidating understanding in a visual-spatial form) to a Gallery Walk (peer teaching and critique). This layered sequence ensures that explanatory ideas are first	Teacher uses three checkpoints: (1) During cold-call discussion: are learners able to generate functional explanations (not just restate structural observations)? A learner who says 'villi increase surface area' meets expectations; a learner who adds 'and the capillary network inside each villus means glucose can immediately enter the bloodstream after crossing the epithelial cells' exceeds expectations. (2) During group annotation: are function statements connected to the

readings	villus cells: greatly increase surface area for absorbing glucose from ugali starch (already digested by amylase and maltase — Lesson 3).' After 8 minutes of group annotation, each group nominates one adaptation for a Gallery Walk presentation (2 minutes per group, posted on the wall).	metres long to absorb the same nutrients from our ugali. Because of villi and microvilli, 6 metres is enough.' Second, ask: 'Our stomach produces hydrochloric acid and pepsin to digest the proteins in nyama choma. Why does the stomach not digest its own lining?' WAIT 15 seconds. Cold-call 2 learners. If no learner mentions mucus, say: 'Think about what goblet cells on your villus diagram might produce — what is their function?' Bridge to the concept of mucus lining and pepsinogen activation. Third, ask: 'What would happen to digestion if the pyloric sphincter opened too quickly and released half-digested chyme into the small intestine before enzymes had finished working?' WAIT 12 seconds. Cold-call 2 learners. After the 3-minute discussion, release groups to annotate for 8 minutes. During the Gallery Walk (10 minutes), each group posts their annotated diagram and one spokesperson presents their chosen adaptation in 2 minutes while other learners use sticky notes or peer-feedback slips to add one supporting detail or one question to each poster. Teacher circulates and asks: 'Show me on your diagram where glucose from the ugali starch would cross into the bloodstream — trace the exact path.'	aired publicly, then applied in writing, then stress-tested by peers. Connecting every annotation back to a specific food component from the Kenyan meal (ugali starch → glucose → villus absorption) anchors the structural knowledge in the phenomenon rather than leaving it as abstract anatomy.	meal? Prompt any group whose annotations say only 'villi absorb nutrients' without specifying which nutrient or which meal component. (3) During Gallery Walk: do sticky-note additions from peer groups demonstrate comprehension or confusion? Collect all sticky notes at the end — this is a rich formative dataset for Lesson 7 planning.
Driving Question Board (DQB) Creation 📹 **VIDEO:** [Centripetal force problem solving] Source: Khan Academy (English	Learners return to the class Driving Question Board — the large display established in Lesson 1 containing the original questions, sticky notes, and the evolving model of digestion. Each group receives three colour-coded cards: GREEN (a question from the DQB that can now be answered with	Direct learners' attention to the DQB. Say: 'We have been building evidence across six lessons. Let us see what today's lesson has unlocked.' Gesture to the original questions on the board and ask: 'Which of these questions — the ones you asked in Lesson 1 when you looked at the ugali and	DQB Curation with Colour-Coded Progress Cards (NGSS Practice 1 — Asking Questions; Practice 6 — Constructing Explanations): The DQB is not a static display but a living tracking tool. The three-colour system makes epistemic progress visible: learners can literally see which questions have	Teacher audits the GREEN cards posted for accuracy: are learners citing the correct evidence? A GREEN card that says 'The stomach does not digest itself because of mucus and pepsinogen — Lessons 3 and 6' demonstrates strong cross-lesson integration. A GREEN card that says 'The body

- US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Surface Area and Nets (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	evidence from Lessons 1–6), YELLOW (a question that is partially answered but needs more evidence), and RED (a new question raised by today's lesson). Groups spend 4 minutes writing their cards and then post them on the DQB. The class briefly reviews the board together (2 minutes), focusing on which original questions — particularly 'How does the body know which chemical to release at each stage?' and 'How does the stomach avoid digesting itself?' — can now be moved to the 'Answered' section.	sukuma wiki — can you now answer using structural evidence from today, plus chemical evidence from Lessons 2, 3, 4, and 5?' Allow 4 minutes for group card-writing. As groups post, read one GREEN card aloud and ask: 'Which lesson gave you the evidence to answer this?' Cold-call the group that posted it. Then say: 'What new questions did today's lesson raise for you?' Read one RED card aloud. Say: 'We will explore some of these new questions in Lesson 7, when we look at what happens when these adaptations fail — for example, when the mucus lining of the stomach breaks down, or when the cardiac sphincter weakens. Keep these questions visible.' WAIT TIME: After asking 'which lesson gave you the evidence?', wait 10 seconds before calling on the group.	moved from 'we don't know' to 'we have evidence.' This metacognitive practice builds scientific habits of mind — specifically the understanding that knowledge accumulates across multiple lines of evidence and that answered questions often generate new, more sophisticated questions.	knows what chemicals to release because of enzymes' without specifying region or trigger demonstrates surface-level understanding. Collect all RED cards — new questions raised — as formative intelligence about conceptual gaps to address in Lessons 7 and 8.
Model Building Phase 📹 **VIDEO:** Centripetal force problem solving Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Surface Area and Nets (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners retrieve the cumulative annotated diagram of the alimentary canal they have been building since Lesson 1. Using a new colour of pen (to show this lesson's additions), they add three structural annotations to their model: (1) a zoomed inset of the small intestinal wall showing villi and microvilli, labelled with surface area function and absorption pathway for glucose (from ugali), amino acids (from nyama choma), and fatty acids (from nyama choma fat); (2) the stomach interior with rugae labelled and an arrow showing the mucus layer; and (3) the cardiac and pyloric sphincters labelled with their regulatory functions. In the margin, each learner writes one sentence answering the question: 'How does	Say: 'Open your cumulative digestive system model — the one you have been adding to since Lesson 1. Take a new colour pen or pencil. You have 4 minutes to add today's structural evidence to your model. I want to see three additions: the villus inset with labels, the rugae and mucus layer in the stomach, and both sphincters with their functions. And write one sentence in the margin — your best current answer to the central puzzle: how does a 3 cm tube absorb everything from ugali, sukuma wiki, and nyama choma?' After 4 minutes, ask one learner to read their margin sentence aloud. Respond with: 'That is a strong explanation. Can anyone add one piece of structural evidence to make it even more complete?'	Cumulative Model Revision with Colour-Coded Layering (NGSS Practice 2 — Developing and Using Models): Using a different colour for each lesson's additions makes the growth of understanding literally visible on the page. The cumulative model is the learner's personal scientific record — a physical embodiment of the storyline. The margin sentence is a metacognitive demand: learners must synthesise structural and chemical evidence into a single coherent explanation, which mirrors the scientific practice of constructing explanations from multiple evidence types.	Teacher circulates during the 4-minute model revision and checks for three non-negotiables: (1) Is the villus inset showing the internal capillary network (not just the outer shape)? This detail is essential for understanding absorption into the bloodstream. (2) Are both sphincters correctly positioned (cardiac at oesophagus-stomach junction, pyloric at stomach-duodenum junction) — not just labelled as 'valves' in a generic location? (3) Does the margin sentence reference surface area AND the specific nutrients from the Kenyan meal (not just 'nutrients' generically)? Collect 4–5 margin sentences by asking learners to read them as an exit check — this determines which structural concepts need reinforcement at

	the structural design of the small intestine solve the puzzle of absorbing everything from our Kenyan meal through a tube only 3 cm wide?'	Cold-call one additional learner. Finally, preview Lesson 7: 'In our next lesson, we will ask a harder question: what happens when these beautiful adaptations fail? What goes wrong when the mucus lining breaks down, or when the pyloric sphincter does not close properly? That is where the real medical science begins.' WAIT TIME: After 'can anyone add one piece of structural evidence?', wait 12 seconds before cold-calling.		the start of Lesson 7.

D. TEACHER REFLECTION
1. During the Explain Phase cold-call discussion, how many learners were able to explain why the stomach does not digest itself (mucus lining + pepsinogen activation) without prompting? What does this reveal about whether Lesson 3's enzyme content was sufficiently retained, and how should I address this gap at the start of Lesson 7? 2. When learners annotated their diagrams during the Gallery Walk, did their function statements connect structural features to specific food components from the Kenyan meal (ugali → starch → glucose → villus absorption), or did they remain generic (e.g., 'villi absorb food')? What scaffolding would help learners make the structure-to-specific-nutrient connection more consistently? 3. Looking at the RED cards from the DQB update, what new questions did learners generate? Are these questions about structural failure (pointing toward Lesson 7 — disorders) or about mechanisms at a deeper level (pointing toward further study)? How can I use the most productive RED cards to open Lesson 7? 4. During the Model Building Phase, how many learners included the capillary network inside the villus in their inset diagram — indicating they understand absorption as entry into the bloodstream, not just crossing the epithelial surface? For those who omitted it, what targeted intervention (a 2-minute sketch demonstration at the start of Lesson 7) would correct this efficiently? 5. Was 40 minutes sufficient for the Gallery Walk (10 minutes including posting and peer sticky-notes) AND the DQB update AND the Model Building Phase? If time was short, which phase was compressed, and how might I restructure the sequence to protect the DQB update — the most metacognitively valuable phase in this EXPLAIN lesson?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners examined labelled diagrams and photomicrographs of three structural features: (1) villi and microvilli (brush border) of the small intestinal wall, showing finger-like projections, internal capillary networks, and lacteals; (2) rugae folds of the stomach wall and the goblet-cell mucus layer; and (3) the cardiac and pyloric sphincters at the entry and exit of the stomach.
What did I learn?	The small intestine's villi and microvilli increase its absorptive surface area by up to 600-fold, allowing a 3 cm wide tube to absorb

	all glucose (from ugali starch), amino acids, and fatty acids (from nyama choma) efficiently. The stomach's rugae allow it to expand and maximise contact with food during churning, while the mucus lining (from goblet cells) and the secretion of pepsin as inactive pepsinogen prevent the stomach from digesting its own wall. The cardiac sphincter prevents stomach acid from refluxing into the oesophagus, while the pyloric sphincter controls the rate at which partially digested chyme enters the small intestine, ensuring enzymes have time to act.
How does this explain the phenomenon?	The puzzle of how a narrow small intestine absorbs all nutrients from a Kenyan meal is resolved by structural adaptation: massive surface area amplification through villi and microvilli means the intestinal lining contacts and absorbs digested nutrients at an extraordinary rate. The stomach's self-protection mechanisms (mucus, pepsinogen) explain why the body does not digest itself despite producing powerful acid and proteases. The sphincters explain how the body 'knows' when to move food from one region to the next — they are mechanical gatekeepers that coordinate timing across the alimentary canal. Together, these structural adaptations answer two of the original DQB questions from Lesson 1 and set up the investigation of what happens when these adaptations fail.

LESSON 7 (40 minutes): Model Building — Making a 3D Model of the Human Digestive System	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners consolidate six lessons of evidence about digestion by constructing a labelled 3D physical model of the alimentary canal that maps every organ, its mechanical/chemical processes, and the enzymes involved — directly answering how the ugali-and-sukuma-wiki meal is progressively broken down and absorbed.
Knowledge	– Name and correctly sequence every major organ of the human alimentary canal (mouth, oesophagus, stomach, small intestine, large intestine, rectum) and the associated glands (salivary glands, liver, pancreas). – State at least one mechanical and/or chemical process that occurs in each region, including the specific enzyme(s) involved and the substrate each enzyme acts upon. – Explain the role of bile (produced by the liver, stored in the gall bladder) in emulsifying fats from nyama choma before lipase can act, and why this is critical for fat digestion.
Skills	– Construct a labelled 3D physical model of the alimentary canal using locally available materials (cardboard tubes, plastic bags, cloth/sponge, string), accurately representing relative organ sizes and connections. – Apply a peer-assessment checklist to evaluate another group's model, providing specific, evidence-based feedback that references lesson evidence from Lessons 2–6.
Attitudes	– Demonstrate creativity and pride in representing a biological system accurately with everyday Kenyan materials, connecting craft to scientific understanding. – Appreciate that the digestive system's complexity — revealed organ by organ across this unit — explains phenomena that initially seemed impossible, such as a 3 cm tube absorbing all nutrients from a full school lunch.
Key Inquiry Question	How is the human digestive system adapted to its functions?
Purpose in Storyline	This is the culminating model-building lesson of the evidence-gathering sequence. Learners have separately investigated mechanical digestion (L2), enzymes by region (L3), food tests (L4–L5), and structural adaptations (L6). Here they synthesise all of that evidence into a single physical artefact — the 3D alimentary canal model — that provides a unified, visual answer to the driving question: how does the body extract nutrients from ugali, sukuma wiki, and nyama choma without visible machinery?
Safety Notes	Scissors and craft knives used for cutting cardboard must be handled with care; demonstrate safe cutting technique before groups begin. No food or drink during model construction. Wash hands after handling materials. Ensure no sharp wire ends are exposed on completed models.

B. LESSON OVERVIEW

Lesson 7 is the designated Model Building lesson in the eight-lesson evidence-gathering sequence for Sub-Strand 1.3. Having spent Lessons 2–6 gathering specific pieces of evidence — about mechanical versus chemical digestion, enzyme specificity, food-test results, and structural adaptations such as villi — learners now face the intellectual challenge of integrating all of that evidence into a single coherent representation: a 3D physical model of the human alimentary canal. The model must label each organ, annotate at least one mechanical and/or chemical process per region, and identify the enzyme(s) involved. Using locally available materials (cardboard tubes cut from paper rolls for the oesophagus and intestines, a small plastic bag for the stomach, damp cloth or sponge strips for villi lining the small intestine, string tied as sphincters, and a larger plastic bag or balloon for the large intestine), groups work collaboratively to build and label their models within a tight 15-minute construction window.

The lesson opens with a brief Predict Phase in which learners sketch from memory the sequence and relative sizes of digestive organs — a retrieval practice exercise that surfaces gaps before construction begins. The Observe Phase is reconceived here as a 'gallery review': groups spend five minutes revisiting their evidence notebooks and the DQB wall, specifically looking for which enzymes, substrates, and adaptations they recorded in earlier lessons, treating these records as the 'data' they must now represent in three dimensions. The Explain Phase is embedded within model construction itself: the teacher circulates with probing questions (e.g., "Show me where bile enters — what does bile do and why is that important for the fat in your nyama choma?") that force learners to articulate the reasoning behind each modelling choice rather than simply assembling materials.

Peer assessment using a teacher-prepared checklist follows construction, giving groups structured feedback before a whole-class share-out. The lesson closes with a DQB update — groups move question cards that their model now answers into the 'Explained' column — and a brief model revision prompt. By the end of the lesson, every group has a tangible, annotated physical model that represents the complete answer to the driving question, ready for the summative lesson (Lesson 8).

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Banking 3: Fractional reserve banking Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Scale Distances or Dimensions: Model Car (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Working individually and in silence for 3 minutes, each learner draws a quick sketch of the alimentary canal from memory — labelling as many organs as they can in the correct sequence and noting beside each organ one process they believe occurs there. Learners then compare their sketches with a partner for 1 minute, circling any organ or process they disagree on. These disagreements become the 'building targets' for the model construction that follows.	Before distributing any materials, say: 'Before we build anything, I want each of you to close your eyes for 10 seconds and picture the journey of a piece of ugali from the moment it enters your mouth to the moment waste leaves your body. Open your eyes — now draw that journey.' Allow 10 seconds of silent thinking (WAIT TIME). Circulate for 3 minutes observing sketches without commenting. After partner comparison, cold-call 3 pairs: 'Tell me one organ you and your partner disagreed on — and why.' Record the three disagreements on the board under the heading 'Test your model against these.' Do NOT resolve the disagreements yet — frame them as predictions to check during construction.	Retrieval Mapping: Learners reconstruct prior knowledge from memory before receiving any scaffolding or materials. This activates and makes visible the mental model each learner currently holds, creating productive cognitive tension between incomplete individual sketches and the complete 3D model they are about to build. It also primes learners to notice exactly which organs or processes they are uncertain about — making them more attentive during construction.	Teacher scans sketches for two key indicators: (1) Does the learner correctly sequence mouth → oesophagus → stomach → small intestine → large intestine → rectum/anus? (2) Does the learner include accessory organs (liver/gall bladder, pancreas) or only the alimentary canal tube? Groups missing accessory organs are mentally flagged for targeted questioning during the construction phase. Disagreements recorded on the board reveal the class's collective uncertainty map.
Observe Phase 📹 **VIDEO:** Banking 3:	Groups spend 5 minutes in a structured 'Evidence Retrieval Gallery' — each learner opens their evidence notebook and the	Distribute building-list template sheets (or instruct learners to rule three columns: Organ | Process(es) | Enzyme +	Evidence Notebook Synthesis: By treating their own lesson records as a primary source — rather than a textbook — learners practise the	Teacher checks building lists for completeness: Does the list include the mouth (salivary amylase, starch → maltose),

[Fractional reserve banking] Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Scale Distances or Dimensions: Model Car (PLIX) Source: CK-12 🔍 Search ARES for similar readings	group nominates one scribe to compile a shared 'building list' on a half-sheet of paper. The list must record: (a) every organ name from Lessons 2–6 evidence, (b) the mechanical and/or chemical process at each organ, and (c) the enzyme name and its substrate. Groups also briefly revisit the DQB wall, identifying which question cards relate to enzyme location or structural adaptation — these are the questions their model must answer. The building list becomes the specification document for model construction.	Substrate). Say: 'You have 5 minutes. Your evidence notebooks and the DQB wall are your only sources. Do not guess — find the evidence you recorded in Lessons 2 through 6. Your building list is your contract: you will only put a label on your model if you can point to the evidence that supports it.' Set a visible timer. After 4 minutes, give a 1-minute warning. Cold-call one group: 'Read me your entry for the small intestine — enzyme, substrate, and one structural adaptation.' (Expected answer should reference maltase/sucrase/peptidase, maltose/sucrose/peptides, and villi from Lesson 6.) Use their answer to briefly validate the evidence-retrieval process before construction begins.	science and engineering practice of 'Obtaining, Evaluating, and Communicating Information.' The building list externalises their collective understanding into a specification document, making implicit knowledge explicit and checkable. This mirrors how engineers create a design brief before fabrication.	stomach (pepsin, proteins → peptides; hydrochloric acid), small intestine (pancreatic amylase, lipase, trypsin/chymotrypsin; bile from liver/gall bladder), and large intestine (water reabsorption, no enzymes)? Any group whose list omits the liver/pancreas as accessory organs is prompted: 'Where does the enzyme that digests the fat in your nyama choma actually come from — is it made inside the small intestine?' (pointing them back to Lesson 3 evidence).
Explain Phase 📹 **VIDEO:** Banking 3: Fractional reserve banking Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Scale Distances or Dimensions: Model Car (PLIX) Source: CK-12 🔍 Search ARES for similar readings	Groups construct their 3D models over 15 minutes using the provided locally available materials. Each organ must be physically represented and carry a label card showing: (a) organ name, (b) mechanical and/or chemical process, and (c) enzyme(s) and substrate(s). String tied around the oesophagus–stomach junction and the stomach–small intestine junction represents the cardiac and pyloric sphincters respectively. Damp sponge or cloth strips glued or folded inside the small intestine tube represent villi. Once construction is complete (or at the 15-minute mark), groups swap models with an adjacent group and spend 4 minutes completing a peer-assessment checklist provided by the teacher, writing specific written feedback on a sticky note attached to the model.	Circulate continuously during construction. Target each group with at least one probing question drawn from the following bank — choose based on what you observe the group building or labelling: – (Stomach) 'Your plastic bag is the stomach — show me the string sphincters. Now tell me: why does the stomach need a sphincter at each end? What would happen to your ugali digestion if the pyloric sphincter didn't exist?' – (Liver/Bile) 'Show me where bile enters the small intestine. What does bile actually do to the fat in the nyama choma — and why does fat need that step before lipase can work?' [WAIT 12 seconds for response before prompting.] – (Villi) 'Touch your sponge villi	3D Artefact Construction with Embedded Explanation: The act of physically building the model forces learners to make consequential decisions — where to tie the string sphincters, how much sponge to use for villi, which label to write — each decision requiring a justification rooted in prior lesson evidence. This is the NGSS Science and Engineering Practice of 'Developing and Using Models.' Peer assessment using a structured checklist then introduces the practice of 'Obtaining, Evaluating, and Communicating Information' — learners must read another group's model as a scientific argument and assess whether the evidence is correctly represented.	Teacher uses the peer-assessment checklist as a live formative tool — collecting the sticky-note feedback after the lesson to identify systemic misconceptions. Key indicators to check: (1) Is bile correctly attributed to the liver (produced) and gall bladder (stored), entering via the bile duct into the duodenum — not produced by the stomach? (2) Are both mechanical AND chemical processes labelled for the mouth (chewing + salivary amylase) and stomach (churning + pepsin + HCl)? (3) Is the large intestine correctly labelled for water reabsorption only, with no digestive enzymes? (4) Are villi represented in the small intestine, not the stomach or large intestine? Groups with errors on (1) or (3) are priority targets for Lesson 8 review.

		— you told me in Lesson 6 that these increase surface area 600-fold. Show me on your model where glucose from the ugali's starch crosses into the bloodstream. What vessel does it enter first?' – (Large intestine) 'Your large intestine has no enzyme labels — is that a mistake or is that correct? Defend your answer.' After peer assessment, facilitate a 3-minute whole-class share-out: cold-call 2 groups to share the most useful piece of feedback they received. Say: 'The best models don't just show what — they show why. Does your model explain why the small intestine is so long, or does it just show that it is long?'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Banking 3: Fractional reserve banking Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Scale Distances or Dimensions: Model Car (PLIX) Source: CK-12 🔍 Search ARES for similar readings	With their completed model in front of them, each group selects two question cards from the DQB wall that their model now provides a visible, labelled answer to. A group spokesperson reads the question aloud, points to the specific part of the model that answers it, and moves the card to the 'We can now explain this' column. Groups also write one new question card — something the model made them realise they still cannot explain — and add it to the 'Still wondering' column. The teacher records all new questions on the board.	Point to the DQB wall and say: 'Since Lesson 1, we have been collecting questions about how the body handles our school lunch of ugali, sukuma wiki, and nyama choma. Your model is now a physical answer machine. Walk to the board, find a question your model answers, and bring it back to your table.' Allow 2 minutes for groups to select cards. Then cold-call 3 groups to present: 'Hold up your model, point to the relevant part, and read the question — then explain how your model answers it.' After presentations, say: 'Is there anything building this model made you MORE confused about — something you couldn't figure out how to represent?' [WAIT 10 seconds.] Accept 2–3 new questions and add them to the board. Explicitly connect to the anchoring phenomenon: 'We started this unit with the puzzle of	DQB Card Migration: Moving question cards from 'Wondering' to 'Explained' is a visible, public record of the class's growing explanatory power. By requiring learners to physically point to the model feature that answers the question, the teacher ensures the migration is grounded in evidence rather than vague confidence. New question cards generated in this phase reveal the productive edges of current understanding — particularly about topics not yet covered (e.g., how the body avoids digesting itself, addressed in Lesson 8).	Teacher tallies how many cards each group successfully migrates to 'Explained' and whether their justifications are accurate. A group that cannot migrate any card, or migrates a card with an incorrect justification, is identified for a brief 1-on-1 check-in during Lesson 8. The new 'Still wondering' cards are photographed (or written into the teacher's notes) to inform the design of Lesson 8's closing activities.

		how a 3 cm tube absorbs everything from a plate of ugali and sukuma wiki. Can someone point to the part of their model that solves that puzzle?' (Expected: villi in the small intestine.)		
Model Building Phase 📹 **VIDEO:** Banking 3: Fractional reserve banking Source: Khan Academy (English - US curriculum) 🔍 Search ARES for similar videos 📖 **HTML:** Scale Distances or Dimensions: Model Car (PLIX) Source: CK-12 🔍 Search ARES for similar readings	In the final 4 minutes, each group makes at least one physical revision to their model based on the peer-assessment feedback they received. The revision must be annotated — the group writes a small 'Revision Note' card attached to the changed part, stating: 'We changed [X] because [evidence from Lessons 1–6].' Groups that finish early are challenged to add a 'Failure Mode' tag to one organ — a red sticky note describing what goes wrong with digestion when that organ or its enzyme fails (e.g., 'If the pancreas stops producing lipase, the fat in nyama choma cannot be digested — it passes into the large intestine unabsorbed, causing steatorrhoea'). Models are displayed on desks for the teacher to photograph for assessment records.	Say: 'Science models are never finished — they are always revised when new evidence arrives. You have 4 minutes to make one improvement to your model based on the feedback your peers gave you. Write a revision note so that anyone looking at your model can see what changed and why.' Circulate to ensure revision notes cite specific lesson evidence, not vague statements like 'it was wrong.' For groups adding Failure Mode tags, prompt: 'Think about what we know from Lesson 3 about enzyme specificity — and from Lesson 6 about villi. If either of those structures fails, what part of our ugali-and-sukuma-wiki puzzle goes unsolved?' [WAIT 15 seconds.] In the final 1 minute, ask the whole class: 'When we started Lesson 1, someone said the stomach digests everything. Hold up your model — where is that idea wrong, and what is the evidence?' Cold-call 2 learners for brief responses.	Iterative Model Revision with Evidence Citations: Requiring a written Revision Note that cites specific lesson evidence transforms model-building from a craft activity into a scientific practice — directly mirroring how scientists revise models when new data arrives. The Failure Mode extension connects structural adaptation (Lesson 6) to clinical consequence, bridging biology and health literacy in a Kenya-relevant way (e.g., pancreatic insufficiency, coeliac disease affecting villi). This is the NGSS practice of 'Developing and Using Models' at its highest level for Grade 10: not just building a model, but evaluating its limits.	Teacher photographs all completed models before the end of the lesson for use in Lesson 8 summative assessment. Key formative indicators on the revision notes: (1) Does the cited evidence reference a specific lesson observation or food-test result, or is it a textbook statement copied from the board? (2) Does the Failure Mode tag (where attempted) correctly link a specific enzyme or structure failure to a specific nutrient that goes undigested? Models are sorted into three categories for teacher follow-up: Complete and Accurate (all five checklist criteria met, revision note evidence-based); Partially Complete (3–4 criteria met, revision note vague); and Needs Reteaching (fewer than 3 criteria met, or systematic errors such as bile produced in the stomach). The latter group receives targeted support in Lesson 8.

D. TEACHER REFLECTION

1. During the Evidence Retrieval Gallery (Observe Phase), did groups draw meaningfully on their evidence notebooks and DQB records, or did they rely primarily on memory or peer copying? What does this tell me about the quality of evidence-recording habits I have cultivated across Lessons 2–6?

2. Which probing question from the Explain Phase bank generated the most productive struggle — and which generated blank stares? What does this reveal about which concepts (bile and emulsification? sphincters? villi?) remain least understood, and how should I prioritise Lesson 8's review?

3. Were the peer-assessment checklist criteria specific enough for Grade 10 learners to give actionable feedback, or did most sticky notes contain vague praise ('good model') rather than evidence-based critique? If the latter, how would I redesign the checklist for next time?

4. How accurately did learners represent the accessory organs — liver, gall bladder, and pancreas — in their models? Did the physical construction constraint (no

obvious 'tube' material for these glands) cause learners to omit them, and if so, what material or scaffold could I provide in a future iteration? 5. Looking at the DQB card migrations: which original Lesson 1 questions remain in the 'Still wondering' column after Lesson 7? Are these questions I expected to leave for Lesson 8, or did learners generate new, more sophisticated questions that go beyond the sub-strand scope — and if so, how do I honour that intellectual growth within the assessment framework?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed — through the act of physical construction — that the alimentary canal is a continuous, regionally specialised tube in which each section has a distinct shape, size, and lining (e.g., thick muscular stomach wall vs. thin, villi-lined small intestine) that directly corresponds to its function. They also observed, via peer-assessment feedback, which organs and processes their classmates struggled most to represent accurately (commonly: accessory organs — liver, gall bladder, pancreas — and the dual role of bile as an emulsifier, not an enzyme).
What did I learn?	Learners consolidated and integrated knowledge from six prior lessons into a single explanatory model: (1) digestion is both mechanical (teeth, stomach churning, peristalsis) and chemical (enzyme-catalysed) and both strategies are needed to fully process ugali, sukuma wiki, and nyama choma; (2) enzymes are region-specific — salivary amylase acts in the mouth on the starch in ugali, pepsin acts in the stomach's acidic environment on the proteins in nyama choma, and pancreatic lipase (after bile emulsification) acts in the small intestine on fats; (3) the small intestine's villi are the structural adaptation that solves the 'invisible machinery' puzzle — massively increasing surface area for absorption without increasing the tube's outer diameter; and (4) sphincters act as valves that control the rate of movement between regions, giving each section time to complete its specific task.
How does this explain the phenomenon?	The 3D model provides a unified physical answer to the anchoring phenomenon: the ugali-and-sukuma-wiki meal does not encounter 'single uniform processing' — instead, it passes through a precisely sequenced series of chemically distinct micro-environments, each releasing the right enzyme at the right pH for the right substrate. The body 'knows' which chemical to release at each stage because each organ's cells are genetically programmed to secrete specific enzymes in response to specific food stimuli (a concept flagged as extending beyond this lesson). Carbohydrates (ugali starch) are partially digested in the mouth and completed in the small intestine; proteins (nyama choma) require the stomach's acid activation of pepsinogen before digestion can begin — explaining why protein digestion takes longer. The villi of the small intestine are the answer to how a 3 cm tube absorbs all nutrients without 'visible machinery': the machinery is microscopic, folded into the wall itself.

LESSON 8 (40 minutes): DQB Synthesis & Assessment — Driving Question Resolution and Unit Wrap-Up	
A. SPECIFIC LEARNING OUTCOMES	
Category	**Specific Learning Outcomes**
Purpose	Learners synthesise all unit evidence to construct a final, complete written answer to the driving question, demonstrate practical and knowledge-based assessment competencies, and close the unit by connecting digestive science to real-world careers in Kenya.
Knowledge	– Describe the complete journey of a Kenyan meal (ugali, sukuma wiki, nyama choma) through every region of the alimentary canal, naming the enzymes, products, and structural adaptations at each stage. – State the four steps of Benedict's test procedure for detecting reducing sugars, and explain what a positive result (brick-red precipitate) indicates about the digested meal. – Match each major digestive enzyme (salivary amylase, pepsin, pancreatic amylase, lipase, trypsin) to its correct organ of secretion, region of action, substrate, and product. – Explain how villi, microvilli, and the rich capillary network of the ileum are structural adaptations for efficient nutrient absorption. – Identify at least one career pathway (e.g., nutritionist, gastroenterologist, food technologist, public health officer, agricultural biochemist) that directly applies knowledge of animal nutrition.
Skills	– Perform a structured food-test practical (Benedict's test, steps 1–4) safely and accurately, recording observations and interpreting results. – Annotate a Lesson 1 journey map using a second colour to show conceptual growth, distinguishing initial predictions from evidence-based explanations. – Write a coherent, evidence-supported collaborative answer to the unit driving question, citing specific mechanisms and Kenyan meal examples. – Evaluate the completeness of the class Driving Question Board by identifying which questions have been fully answered, partially answered, or remain open.
Attitudes	– Demonstrate intellectual honesty by openly acknowledging how their Lesson 1 predictions were incomplete or incorrect, and valuing the process of revision through evidence. – Appreciate how understanding digestive system adaptations has direct relevance to personal health, food choices, and career opportunities in Kenya.
Key Inquiry Question	How is the human digestive system adapted to its functions — and how completely can we now answer this question using all the evidence we have gathered?
Purpose in Storyline	This final lesson closes the investigative arc that began with the Ugali and Sukuma Wiki Puzzle in Lesson 1. Learners have gathered evidence across seven lessons — from mechanical and chemical digestion, to enzyme action, food tests, structural adaptations, and 3D model building. Lesson 8 brings all those evidence threads together: learners annotate their original thinking, collaboratively write the definitive answer to the driving question, complete a practical and written assessment, and connect their learning to careers — transforming the initial cognitive dissonance of the anchoring phenomenon into confident, evidence-based understanding.
Safety Notes	Benedict's test involves boiling water (hot water bath at ~80–90 °C). Learners must use test-tube holders or tongs when placing or removing test tubes from the water bath. No direct flame contact with test tubes containing Benedict's solution. Broken glassware must be reported immediately to the teacher and disposed of in the designated sharps bin. Learners with heat sensitivity should be seated away from the heat source. Ensure the lab has a working first-aid kit available.

B. LESSON OVERVIEW

Lesson 8 is the culminating lesson of the Nutrition in Animals unit and serves as the resolution point of the entire investigative storyline. When the Ugali and Sukuma Wiki Puzzle was first presented in Lesson 1, learners could not explain how the body knew which chemical to release at each stage of digestion, why carbohydrates and proteins took different amounts of time to digest, or how the body avoided digesting itself. Over the course of seven lessons, learners have gathered layered evidence: they distinguished mechanical from chemical digestion, mapped enzyme action region by region along the alimentary canal, performed food tests for starch, reducing sugars, proteins, and fats, analysed the structural adaptations of the small intestine, and built a 3D model of the complete digestive system. Lesson 8 calls all of that evidence forward and puts it to use.

The lesson opens with a Predict Phase in which learners silently return to their Lesson 1 journey maps and original Driving Question Board (DQB) entries, reading their own prior thinking before instruction begins — a metacognitive move that makes conceptual growth visible and personally meaningful. The Observe Phase is a structured practical assessment: learners perform the four-step Benedict's test on a simulated digested meal sample (a glucose solution representing digested ugali starch), following the KICD-mandated procedure. This grounds the final lesson in the same hands-on, evidence-gathering orientation that has characterised the entire unit. The Explain Phase is the intellectual centrepiece: learners work in pairs to co-author a written response to the driving question, then share key sentences in a whole-class synthesis. The DQB phase closes the board — learners physically move question cards to the "answered" column, flag any genuinely unresolved questions as open scientific problems, and celebrate the growth of their collective understanding. The Model Building Phase is a final annotated comparison: learners place their Lesson 1 journey map beside their Lesson 7 3D model and, in a second colour, annotate every mechanism they can now explain that they could not explain at the start.

Assessment in this lesson is intentionally multi-modal, reflecting KICD's emphasis on competency-based assessment: the Benedict's test practical addresses the procedural rubric indicator verbatim ("Correctly performs experiment on test for reducing sugars following all the steps in the procedure"), the enzyme-region written component addresses knowledge SLOs (a), (b), and (c), and the DQB annotation addresses metacognitive and communication competencies. The career connection task at the close of the lesson connects the sub-strand to the broader General Science essence statement — preparing learners for further education and the world of work.

C. LESSON IMPLEMENTATION FRAMEWORK

Phase	Learner Experience	Teacher Moves	Sensemaking Strategy	Formative Assessment Strategy
Predict Phase 📹 **VIDEO:** Digestive System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Human Digestive System (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners receive their original Lesson 1 journey maps (distributed by the teacher before the lesson begins). They read their own Lesson 1 writing silently for 2 minutes, then use a pen of a DIFFERENT colour (provided by the teacher — ideally green or red) to annotate directly on the map: circling statements they now know are correct, crossing out statements they now know are incomplete or wrong, and writing a one-sentence 'upgrade' next to each crossed-out statement. They do this individually and silently. Learners then predict, in writing at	Before the lesson, distribute Lesson 1 journey maps face-down on desks. Provide each learner with a green or red pen. Say: 'Do not turn the map over yet.' Open with the anchoring phenomenon: 'Cast your mind back to the very first lesson. We placed ugali, sukuma wiki, and nyama choma on the table and asked: how does the body know what to do with each of these? You wrote what you thought then. Today, you will read your own past thinking — and judge it honestly.' Pause 5 seconds. 'Now turn over your maps. Read everything you wrote.	Metacognitive Annotation ('Before and After Thinking Map'): By reading and annotating their own prior thinking with a second colour, learners make their conceptual growth physically visible on a single page. This strategy externalises the difference between pre-instruction intuition and post-instruction understanding, making the learning journey concrete rather than abstract. It also activates prior knowledge across all seven lessons simultaneously, priming retrieval for the assessment and synthesis tasks that follow.	Teacher circulates and looks for: (1) Are learners crossing out statements, or leaving everything unchanged? — if no crossings, probe with 'Is everything you wrote in Lesson 1 still correct?' (2) Do upgrade sentences reference specific mechanisms (e.g., 'salivary amylase,' 'villi,' 'hydrochloric acid') or remain vague? — vague upgrades signal surface-level retention. (3) Do the 3 cold-called learners' upgrades reflect evidence from specific lessons (Lessons 2–7)? Flag any learner who cannot produce a single upgrade sentence for

	the bottom of the map, what they think the single most important adaptation of the human digestive system is — before any class discussion.	Do not change anything yet — just read.' Allow 2 minutes of silent reading. Then say: 'Now pick up your coloured pen. Circle every statement you now know is scientifically correct. Cross out every statement that is incomplete or wrong. Next to each cross, write one sentence — just one — explaining what you now know instead. You have 3 minutes.' Monitor the room; do NOT intervene unless a learner is confused about the task. After 3 minutes, cold-call 3 learners: 'Amina — read us one statement you crossed out and tell us your one-sentence upgrade.' WAIT TIME: 12 seconds minimum after each name before prompting. After 3 shares, say: 'At the bottom of your map, write your answer to this question — in one sentence: what is the single most important adaptation of the human digestive system, and why?' Allow 2 minutes. Do not collect responses yet — they will be revisited in the Model Building phase.		targeted support during the Explain phase.
Observe Phase 📹 **VIDEO:** Digestive System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Human Digestive System (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners conduct the KICD-mandated Benedict's test practical as a structured assessment activity. Working in pairs at lab benches, they receive: a test tube containing approximately 2 cm^3 of a simulated 'digested ugali' sample (a dilute glucose solution prepared by the teacher), a second test tube of Benedict's solution, a water bath (a beaker of hot water on a tripod and gauze, pre-heated to ~85°C by the teacher before the lesson), a test-tube holder, and a results observation sheet. They follow the four mandated steps exactly: Step 1 — Put about 2 cm^3 of test solution in a test tube. Step 2 —	Set up water baths before learners arrive — the practical window is tight (8 minutes). As learners settle, say: 'This is your practical assessment for the unit. You will work in pairs, but each person records results independently on their own sheet. The procedure is the four steps you know from Lessons 4 and 5 — I want to see each step done correctly and safely.' Demonstrate Step 3 once — 'Use the holder like this. Do NOT hold the test tube with your bare hand over the hot water bath. Do NOT bring the mouth of the test tube toward your face.' Circulate during the practical. Do	Evidence-to-Mechanism Bridging via Practical Assessment: The Benedict's test is not merely procedural here — it is positioned explicitly as evidence that connects back to the anchoring phenomenon. The glucose detected IS the end-product of the starch digestion of ugali that learners first observed as a 'puzzling white lump' in Lesson 1. By performing the test themselves and then answering 'why does this solution represent digested ugali?', learners bridge observable chemical evidence (colour change) to the invisible enzymatic mechanism (salivary amylase →	Assessment indicators (aligned to KICD rubric): EXCEEDS — learner follows all four steps in correct sequence, uses holder safely, records a brick-red/orange precipitate, AND correctly explains in writing that digested ugali contains glucose (not starch) because amylase has broken the glycosidic bonds in starch, AND names salivary amylase (mouth) and pancreatic amylase (small intestine) as responsible enzymes. MEETS — correctly follows all four steps, observes and records colour change, correctly names glucose as the product. APPROACHES — follows 3 of 4

	Add an equal amount of Benedict's solution and mix. Step 3 — Place the test tube with the mixture in a hot water bath for a few minutes. Step 4 — Observe and record all colour changes. Learners record results independently on their observation sheets, writing the colour change observed and interpreting what it indicates about the glucose present in the digested sample. They then answer two written questions on the sheet: (a) Why does the test solution represent digested ugali rather than whole ugali starch? and (b) Name the enzyme and the organ of action responsible for converting ugali starch into the glucose detected by this test.	NOT tell learners what colour to expect — observe independently. At the 5-minute mark, say: 'If your water bath is working correctly, you should be seeing a change by now. If you see NO change, check: did you mix after adding Benedict's solution? Is the test tube actually in the water?' Cold-call 2 pairs to report results aloud: 'Kioni and Baraka — what colour did you observe and what does it tell you?' WAIT TIME: 10 seconds. After results are shared, say: 'Now answer the two written questions on your sheet. These are part of your assessment — answer independently.' Allow 3 minutes for written questions. Collect observation sheets.	maltose → glucose via maltase) that produced it. This transforms the practical from a skills exercise into a sensemaking event.	steps; correctly records the colour change. BELOW — follows fewer than 3 steps, or cannot record/interpret the colour change. Flag BELOW-level learners for peer support during the Explain phase.
Explain Phase 📹 **VIDEO:** Digestive System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Human Digestive System (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	Learners now write the collaborative final answer to the driving question. The teacher displays the full driving question on the board: 'How is the human digestive system adapted to break down and absorb the nutrients in a typical Kenyan meal — and what happens when those adaptations fail?' Working in pairs (same pairs as the practical), learners have 6 minutes to draft a written answer of 5–8 sentences. The answer must: (1) reference ugali, sukuma wiki, and nyama choma by name; (2) name at least three specific adaptations and explain HOW each adaptation serves a function; (3) name at least two enzyme-region matches with correct substrates and products; (4) include one sentence on what happens when an adaptation fails (e.g., lactase deficiency, ulcers caused by H. pylori disrupting the stomach's mucus lining). Each pair then shares their single strongest	Transition from the practical: 'Put your observation sheets to one side. Now comes the most important 8 minutes of this unit. You have spent eight lessons investigating one question. It is time to answer it — in your own words, using your own evidence.' Display the driving question in large font. Say: 'Work with your partner. Write 5 to 8 sentences. Your answer MUST mention ugali, sukuma wiki, and nyama choma. It MUST name at least three adaptations. It MUST include at least two enzyme-region matches. And it MUST say something about what happens when an adaptation fails — because that was part of the original question.' Set a visible timer for 6 minutes. Circulate — do NOT correct grammar; DO prompt vague statements: 'You wrote "the small intestine absorbs nutrients" — HOW does it do that? What structure makes that possible?' or 'You wrote "enzymes break down	Collaborative Scientific Writing with Evidence Anchoring: Requiring learners to write sentences that explicitly name ugali, sukuma wiki, and nyama choma — the anchoring phenomenon's components — prevents abstract summarising and forces a return to the original concrete context. The 'single strongest sentence' share-out creates a class-authored canonical explanation, which is more powerful than any individual summary because it represents the collective sensemaking of the entire class. Scribing on the board models scientific communication and allows learners to see their own language treated as authoritative — a high-agency, high-accountability sensemaking move.	As pairs write, teacher looks for: (1) Are specific adaptations named (villi, microvilli, mucus lining, teeth, cardiac/pyloric sphincters, coiling of small intestine) or only generic statements? (2) Are enzyme-region pairs correct (e.g., 'salivary amylase in the mouth breaks starch into maltose' — NOT 'enzymes break food')? (3) Does the 'adaptation failure' sentence reference a real condition (ulcer, lactose intolerance, coeliac disease) or remain hypothetical? During the share-out: listen for any pair that uses the word 'adapted' but cannot explain the mechanism — this is the key conceptual gap to address in the DQB phase. Cold-call at least one pair that appeared to struggle during writing to check for understanding before the lesson closes.

	sentence — the one they are most proud of — with the class. The teacher scribes these sentences on the board to build a class 'collective answer.' Learners add any sentences from the board that they did not include in their own response.	food" — which enzyme? Where? Breaking down what?' After 6 minutes: 'Time. Each pair — read me your single strongest sentence. Just one.' Cold-call 5–6 pairs. WAIT TIME: 12 seconds after each pair name. Scribe key sentences on the board verbatim. After 5–6 contributions, say: 'Look at what we have built together on that board. That is a complete, evidence-based answer to the question we asked in Lesson 1. Add any sentence from the board that strengthens your own answer.' Allow 90 seconds for individual additions. Say: 'The body does not accidentally digest ugali, sukuma wiki, and nyama choma. It is exquisitely adapted to do so — region by region, enzyme by enzyme, structure by structure. You now know how.'		
Driving Question Board (DQB) Creation 📹 **VIDEO:** Digestive System Structure and Function - Overview Source: CK-12 🔍 Search ARES for similar videos 📖 **HTML:** Human Digestive System (Flexbook) Source: CK-12 🔍 Search ARES for similar readings	The class returns to their physical DQB (the board or wall display maintained since Lesson 1). Each learner receives two small sticky notes. On the FIRST sticky note, they write one question from the original DQB that they can now fully answer — and write the answer in two sentences on the same note. On the SECOND sticky note, they write one question (from the original DQB, or a new question that has arisen during the unit) that CANNOT yet be fully answered — a question that points beyond this unit. Learners physically move their 'answered' sticky notes from the 'Questions' column to the 'Answered' column on the DQB. The 'unanswered' notes are placed in a new column labelled 'Still Wondering / Open Questions.' The class counts: how many questions	Distribute two sticky notes per learner. Say: 'The Driving Question Board has been our unit's memory since Lesson 1. Every question we posted there was real — something we genuinely did not know. Now we are going to close the board together.' Pause 3 seconds. 'On your FIRST note — pick ONE question from the board that you can now fully answer. Write the question. Write your answer in two sentences. Then move that note to the ANSWERED column.' Allow 2 minutes. Then: 'On your SECOND note — write one question you STILL cannot fully answer, or a new question this unit has opened up for you. This is your intellectual honesty note. Science is full of open questions — it is not a weakness to have them.' Allow 90 seconds. Once notes are placed: count the	DQB Synthesis and Closure ('Answered / Still Wondering' Sort): The physical act of moving question cards from 'Questions' to 'Answered' externalises and celebrates epistemic progress — it makes the growth of the class's collective knowledge visible in a way that a written summary cannot. The 'Still Wondering' column is equally important: it positions remaining uncertainty as intellectually productive rather than as failure, modelling the open-ended nature of scientific inquiry. Connecting unanswered questions to real Kenyan scientists (KEMRI researchers, clinical nutritionists at Kenyatta National Hospital) situates the class's questions within a living scientific community.	Teacher looks for: (1) Are 'answered' notes accompanied by specific, mechanistic answers — or only vague restatements of the question? (2) Do 'still wondering' notes reflect genuine extension thinking (e.g., 'How does the immune system in the gut decide what is food and what is pathogen?' or 'Does the digestive system of a person eating only githeri every day adapt differently over time?') or simply repeat questions that were already answered in the unit? (3) Is the ratio of answered to unanswered questions consistent with the lesson sequence? — a class with fewer than 60% of original DQB questions in the 'Answered' column warrants a teacher note to revisit specific sub-topics in the next unit's spiral review.

	total? How many answered? The teacher leads a brief 3-minute whole-class discussion on which unanswered questions might belong to Grade 11 Biology, which belong to medical science, and which might never be fully answered. The DQB is then photographed by the teacher as a record of unit-level metacognitive growth.	columns together — 'How many questions did we start with? How many are answered?' Cold-call 2 learners to read their unanswered questions aloud. WAIT TIME: 10 seconds. For each, ask: 'Is this a question for Grade 11 Biology, for medicine, or is it an open research question?' Allow brief class responses. Close with: 'The fact that we still have questions is not a failure — it is evidence that we are thinking like scientists. Dr. Kizito Gona at the Kenya Medical Research Institute has spent his career answering questions just like the ones on your second sticky note.' Photograph the board. Say: 'This board shows exactly how far we have come since that plate of ugali and sukuma wiki in Lesson 1.'		
Model Building Phase 📹 **VIDEO:** [Digestive System Structure and Function - Overview] Source: CK-12 🔍 [Search ARES for similar videos] 📖 **HTML:** [Human Digestive System (Flexbook)] Source: CK-12 🔍 [Search ARES for similar readings]	Learners retrieve their Lesson 7 3D model (or the labelled diagram they produced alongside it). They place their annotated Lesson 1 journey map beside the model. Using their second-colour pen, they add a final annotation layer to the journey map — drawing arrows from each crossed-out Lesson 1 statement to the correct mechanism on the model, physically linking the 'before' misconception to the 'after' evidence-based explanation. Where the 3D model is too large to annotate directly, learners annotate the journey map with a model reference code (e.g., 'see model label 6 — villi'). Each learner then writes, at the top of the journey map: one final sentence completing the stem: 'The human digestive system is adapted to break down and absorb the nutrients in ugali, sukuma wiki,	Say: 'We are now going to do the final thing a scientist always does after an investigation — we are going to compare our model at the start with our model at the end, and be honest about what changed.' Distribute Lesson 1 journey maps if not already in learners' hands. Say: 'Place your journey map next to your model or diagram from Lesson 7. With your coloured pen, draw an arrow from each statement you crossed out on your journey map to the correct part of your model. You are literally connecting your old thinking to your new evidence.' Allow 3 minutes for annotation. Then: 'Now — without looking at any notes, any book, or any other person's work — write one sentence at the top of your journey map that begins: The human digestive system is adapted to break down and absorb the nutrients in ugali,	Model Comparison and Final Explanatory Writing ('L1 vs. Final Model'): Placing the Lesson 1 journey map physically beside the Lesson 7 3D model creates a spatial and temporal contrast that makes conceptual growth undeniable. The arrow-drawing task is not decorative — it requires learners to identify the specific mechanism that replaces each original misconception, which is the most demanding form of conceptual revision. The final sentence stem ('…because…') requires causal rather than descriptive language, ensuring that the closing moment of the unit demands mechanism-level thinking rather than mere recall. The career connection task operationalises the KICD General Science essence statement — demonstrating that digestive knowledge has direct application in	Collect journey maps as the primary exit artefact. Teacher assesses: (1) Final sentence quality — does 'because…' introduce a mechanism (enzyme action, structural adaptation, selective absorption) or a description ('because the stomach breaks food')? Mechanistic sentences = meets/exceeds; descriptive only = approaches; no causal language = below. (2) Arrow annotations — do arrows connect specific misconceptions to specific model labels, or are they drawn arbitrarily? (3) Career connection — is the career plausible and is the knowledge connection specific (e.g., 'A nutritionist at Gertrude's Children's Hospital in Nairobi uses knowledge of lactase enzyme deficiency to advise parents of lactose-intolerant children') or generic ('doctors need to know

	and nyama choma because…' — written entirely from memory, without referring to notes. Finally, learners identify ONE career connected to the sub-strand and write a 2-sentence explanation of the connection: career name, and specifically how knowledge of digestive system adaptations is used in that career in Kenya.	sukuma wiki, and nyama choma because…' WAIT TIME after instruction: allow 2 minutes of silent independent writing. Cold-call 3 learners to read their sentences aloud. WAIT TIME: 12 seconds each. After sharing, say: 'Each of those sentences would have been impossible for you to write in Lesson 1. That is what learning looks like.' Final task: 'On the back of your journey map, write one career that uses this knowledge. Name the career. Then write exactly how knowledge of digestive system adaptations is used in that job in Kenya. You have 2 minutes.' Examples the teacher may prompt (only if learners are completely stuck): nutritionist at a Nairobi hospital advising patients with Crohn's disease; food technologist at a uji flour company in Eldoret adjusting starch content for infant formula; veterinary scientist at the University of Nairobi comparing animal and human digestion; gastroenterologist at Kenyatta National Hospital diagnosing ulcers. Collect journey maps as exit artefacts for teacher assessment.	Kenya's economy, health system, and food industry.	about digestion')? Specific + plausible = meets/exceeds. Use annotated journey maps as portfolio evidence of unit-level learning growth for each learner's KICD competency-based assessment record.

D. TEACHER REFLECTION

1. When learners annotated their Lesson 1 journey maps, what proportion crossed out statements versus left everything unchanged? What does this tell you about the depth of conceptual change across the unit — and which specific misconceptions (e.g., 'the stomach digests all food,' 'enzymes are the same in every part of the body') proved most resistant to revision?

2. During the Benedict's test practical assessment, how many pairs followed all four steps in the correct sequence without prompting? For pairs that skipped or misordered steps, was the error procedural (unfamiliarity with the technique) or conceptual (not understanding WHY each step is necessary)? How will you address this in the next food-test unit?

3. In the collaborative driving question writing, did learners' sentences reference specific mechanisms (villi, specific enzymes, mucus lining) or remain at the level of

general statements ('the digestive system is adapted to absorb nutrients')? If general statements dominated, at what point in the unit — Lesson 3 (enzyme-region mapping), Lesson 6 (structural adaptations), or the model-building phase — did deep mechanistic understanding appear to stall?

4. Looking at the DQB 'Still Wondering' column: what proportion of unanswered questions reflect genuine extension thinking that points toward Grade 11 Biology or medical science, versus questions that were actually addressed in the unit but not fully retained by learners? What does this tell you about pacing and the depth of engagement during the Observe and Explain phases of earlier lessons?

5. Did the career connection task reveal gaps in learners' awareness of science-based career pathways in Kenya? Did learners tend to name only medical careers (doctor, nurse) rather than the broader range (food technologist, agricultural biochemist, public health officer, nutritionist)? If so, how might you integrate career awareness more explicitly into individual lesson phases — rather than only in the final lesson — across the next unit?

E. SUMMARY TABLE PROMPT (pre-filled example for this lesson)	
What did I observe?	Learners observed a brick-red/orange colour change when Benedict's solution was added to a simulated digested-ugali sample (glucose solution) and heated in a water bath — providing direct chemical evidence that the starch in ugali has been broken down into reducing sugars (glucose) by the action of amylase enzymes during digestion.
What did I learn?	The human digestive system is structurally and chemically adapted at every region to process the specific nutrients in a Kenyan meal: mechanical digestion (teeth, stomach churning) and chemical digestion (region-specific enzymes — salivary amylase, pepsin, pancreatic amylase, lipase, trypsin) break ugali starch, sukuma wiki fibre, and nyama choma protein into absorbable monomers; villi and microvilli of the ileum (surface area increased up to 600-fold) absorb these monomers into the bloodstream; and protective adaptations (mucus lining, HCl regulation, sphincters) prevent self-digestion and control the rate of passage.
How does this explain the phenomenon?	The Ugali and Sukuma Wiki Puzzle is now fully explained: (1) The body 'knows' which chemical to release at each stage because digestion is controlled by a precise, region-specific enzyme system — each enzyme is secreted only where its substrate arrives, triggered by hormonal and neural signals; (2) Carbohydrates digest faster than proteins because salivary and pancreatic amylase act on the relatively simple glycosidic bonds of ugali starch from the moment the meal enters the mouth, while proteins in nyama choma require the more complex, sequential action of pepsin (acidic, stomach), trypsin and chymotrypsin (alkaline, small intestine); (3) The body avoids digesting itself through mucus lining the stomach and intestinal walls, secretion of inactive enzyme precursors (zymogens) that activate only in the correct environment, and the alkaline mucus secreted by Brunner's glands in the duodenum that neutralises stomach acid before it contacts the intestinal wall.

DIFFERENTIATION AND INCLUSION		
Learner Need	**Support Strategies**	**Extension Strategies**
English Language Learners	Provide bilingual glossaries; allow use of first language for initial model drawing.	Write extended explanations of observations; lead group discussions in English.
Students with learning difficulties	Provide structured note frames; pair with a supportive partner during investigations.	Mentor peers; create additional visual models to explain concepts.
Gifted and advanced learners	Access to additional reading materials; challenge questions during DQB.	Lead model-building discussions; research and present extension topics to class.